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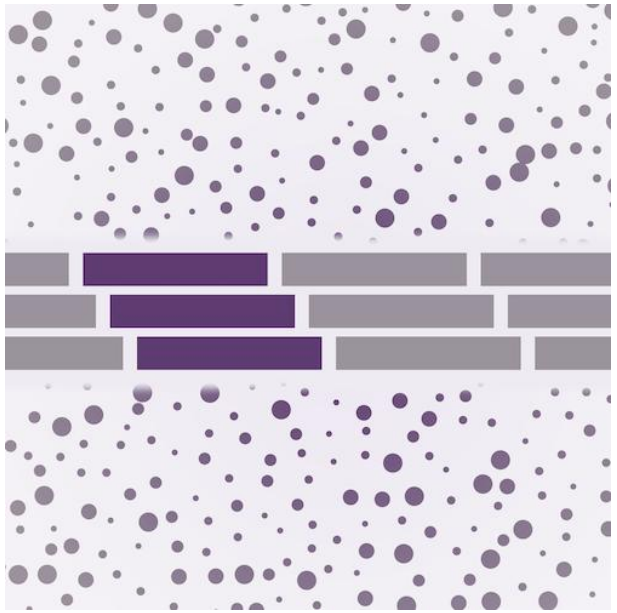
Canadian Bioinformatics Workshops

Module 2: Genome Visualization

Dr. Erin Pleasance

CBW High-Throughput Genomics

November 4-6, 2026



CANADA'S MICHAEL SMITH GENOME SCIENCES CENTRE

Some slides adapted from Rob Syme and previous instructors

Learning Objectives

- Understand the value and diversity of data visualization
- Gain experience with genome browsers
- Explore variation inspection

Overview

- Part I (lecture)
  - Discussion of visualization goals and approaches
  - Overview of IGV
  - Overview of UCSC

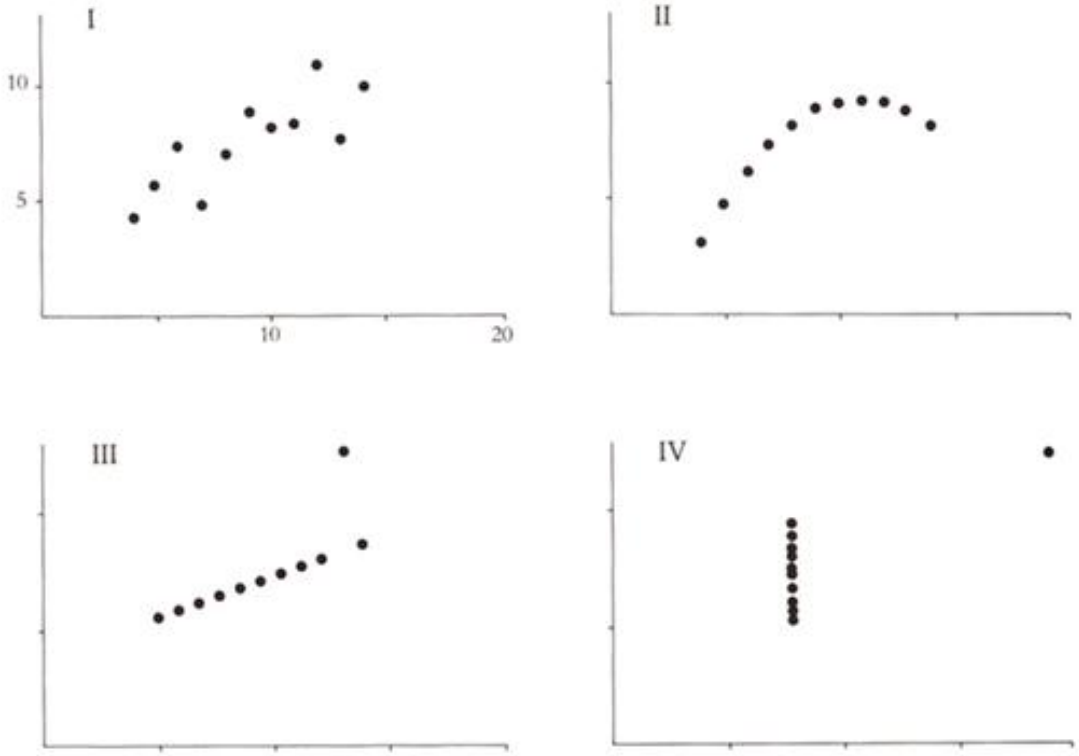
- Part II (lab)
  - Load data and references in IGV
  - Visualize variants in IGV
  - Explore UCSC tutorial



Why visualize?

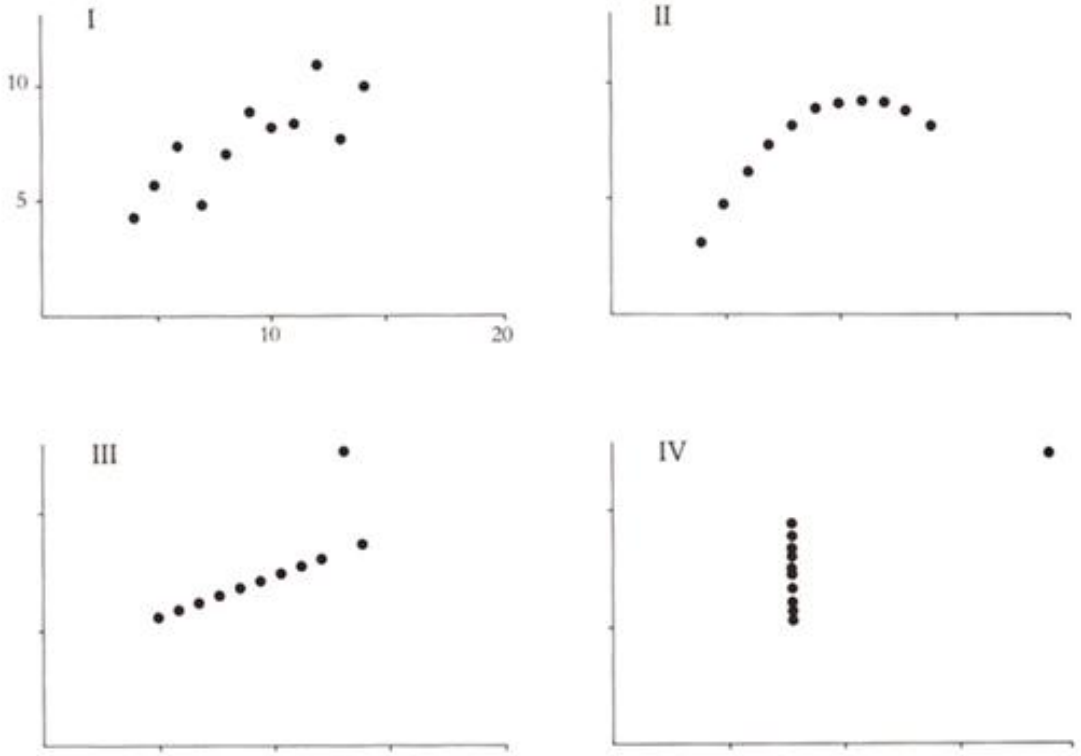
Data interpretation

- Summary statistics don’t tell the whole story



Data interpretation

- Summary statistics don't tell the whole story



anscombe's quartet

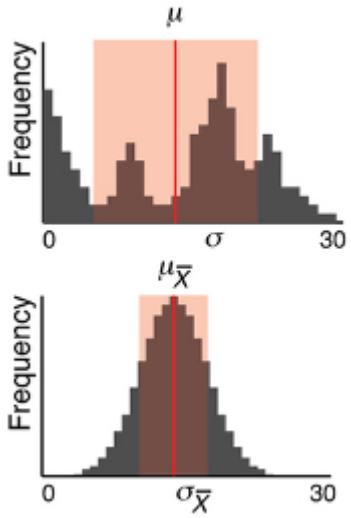
each of the values below is the same for each set

number of points
average x
average y
regression line
standard error of slope
sum of squares
residual sum of squares
correlation coefficient
r^2

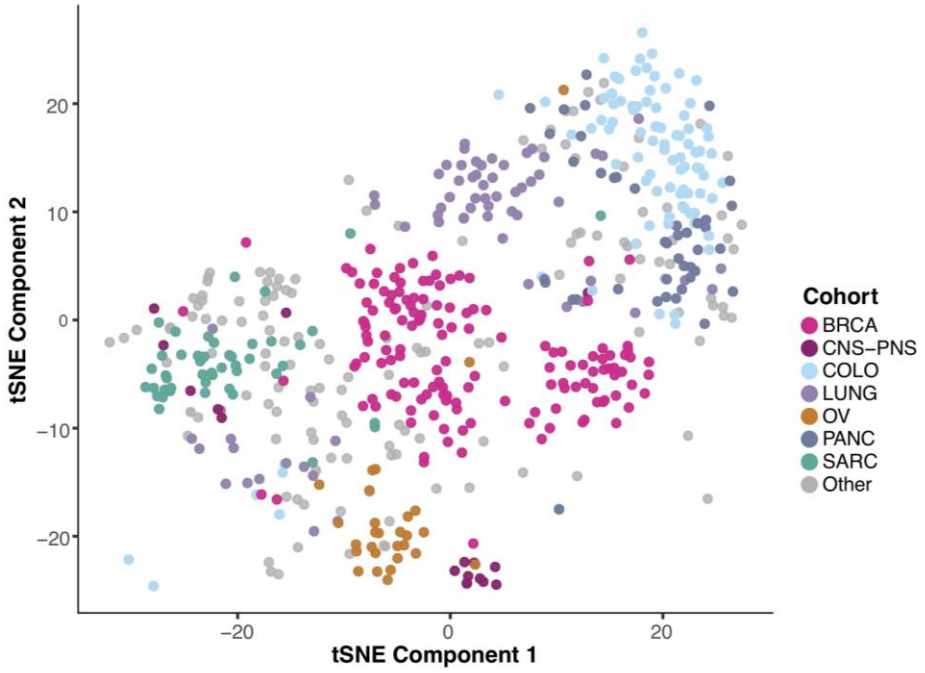
https://mk.bcgsc.ca/talks/datavisualization/

Data interpretation

- How is the data distributed?
- Are there outliers?
- Are there batch effects?
- Does the data make sense?



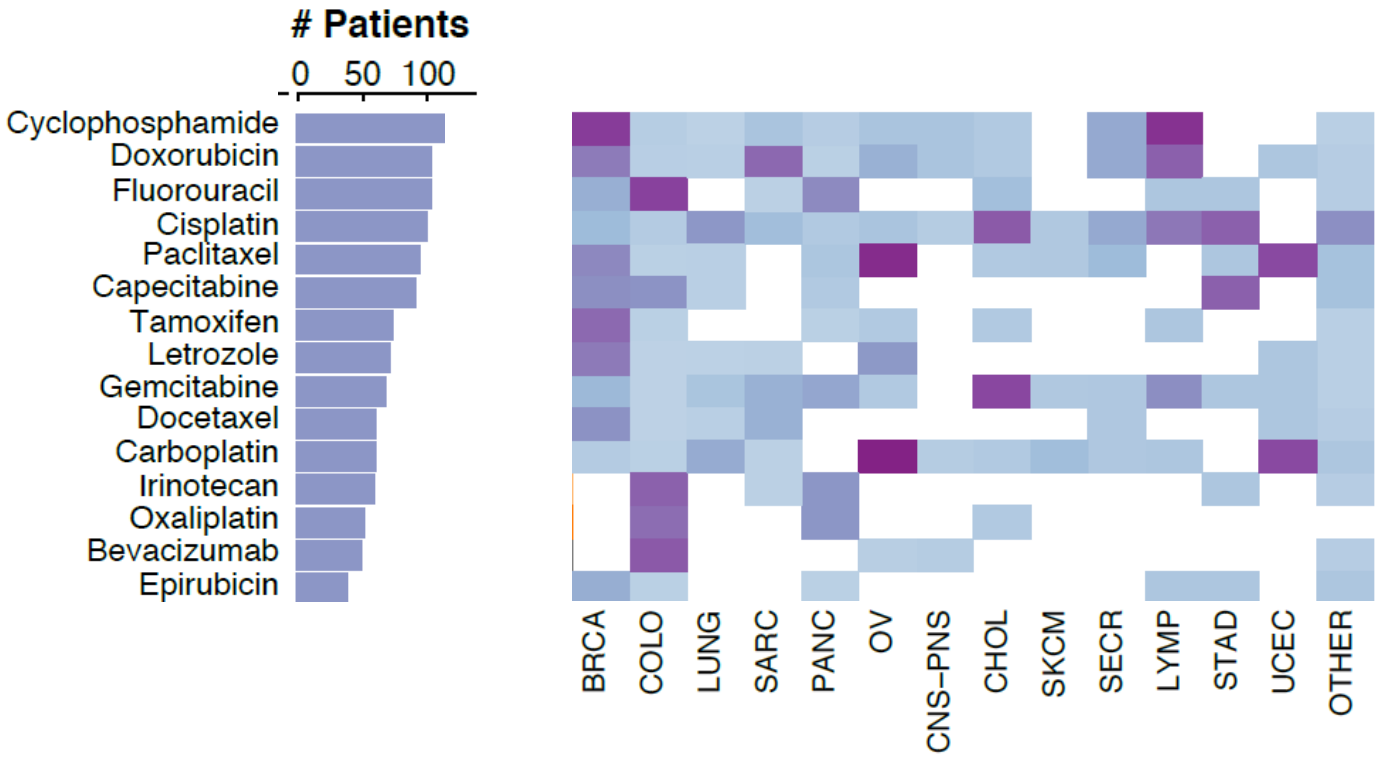
https://blogs.nature.com/methagora/2013/08/giving_statistics_the_attention_it_deserves.html

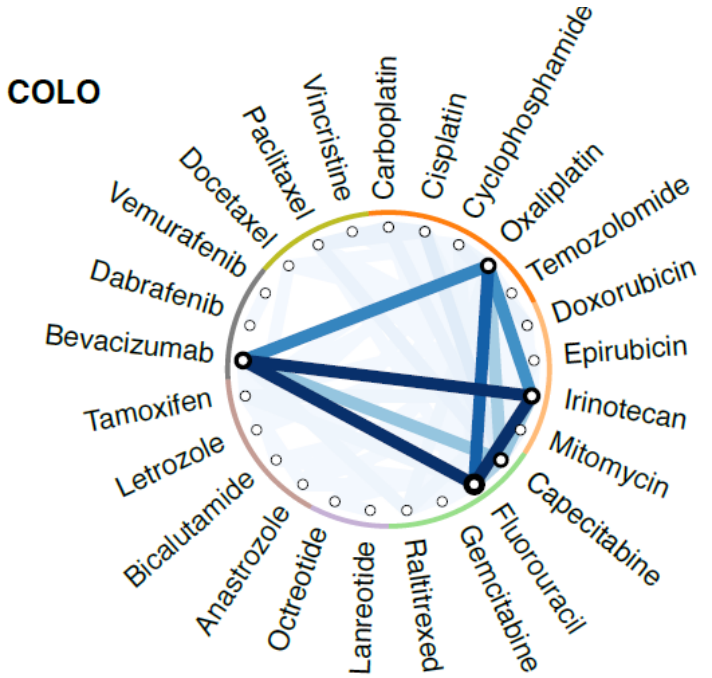


PMID: 35121966

Data size & complexity

- Understanding, finding patterns and relationships in large datasets
- Look at the data multiple ways

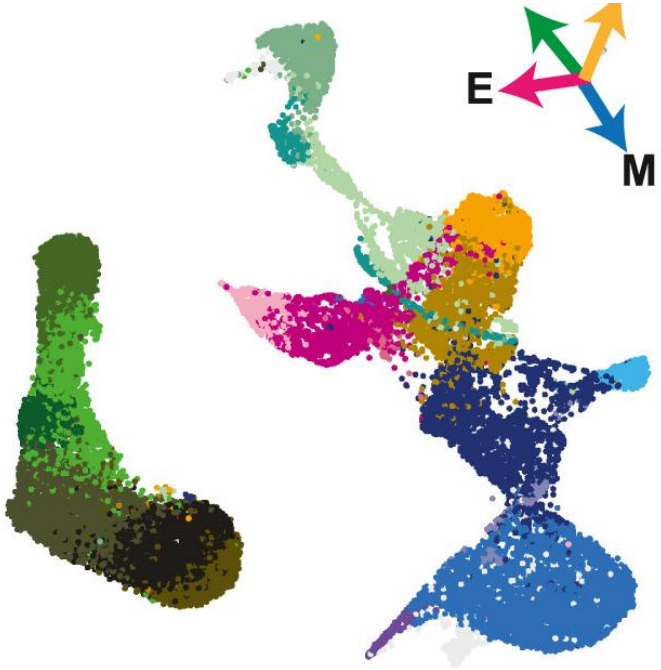




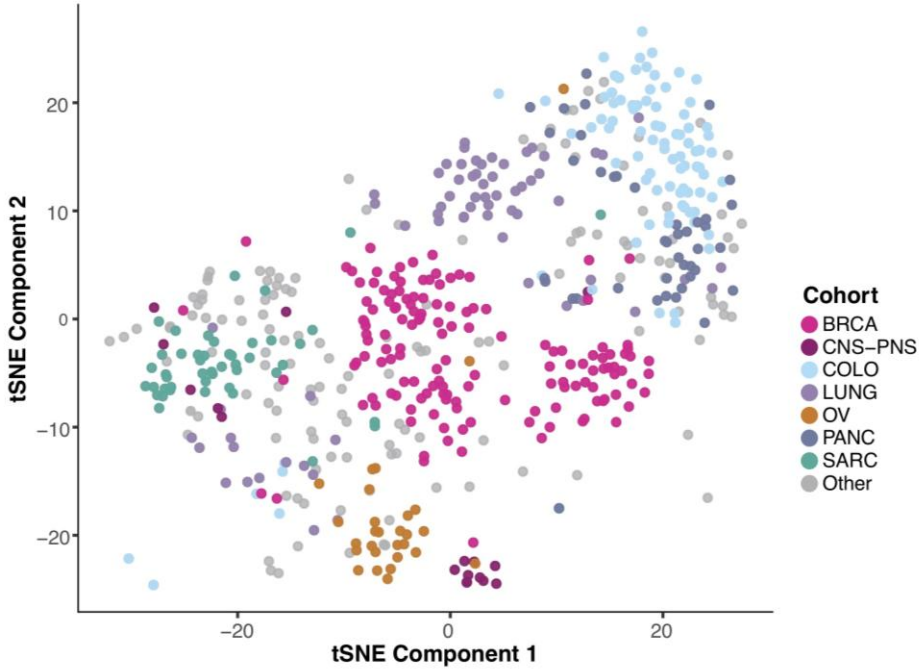
PMID: 35121966

PCA/tSNE/UMAP

- Can use eg. R ggplot2, PCAtools, Rtsne



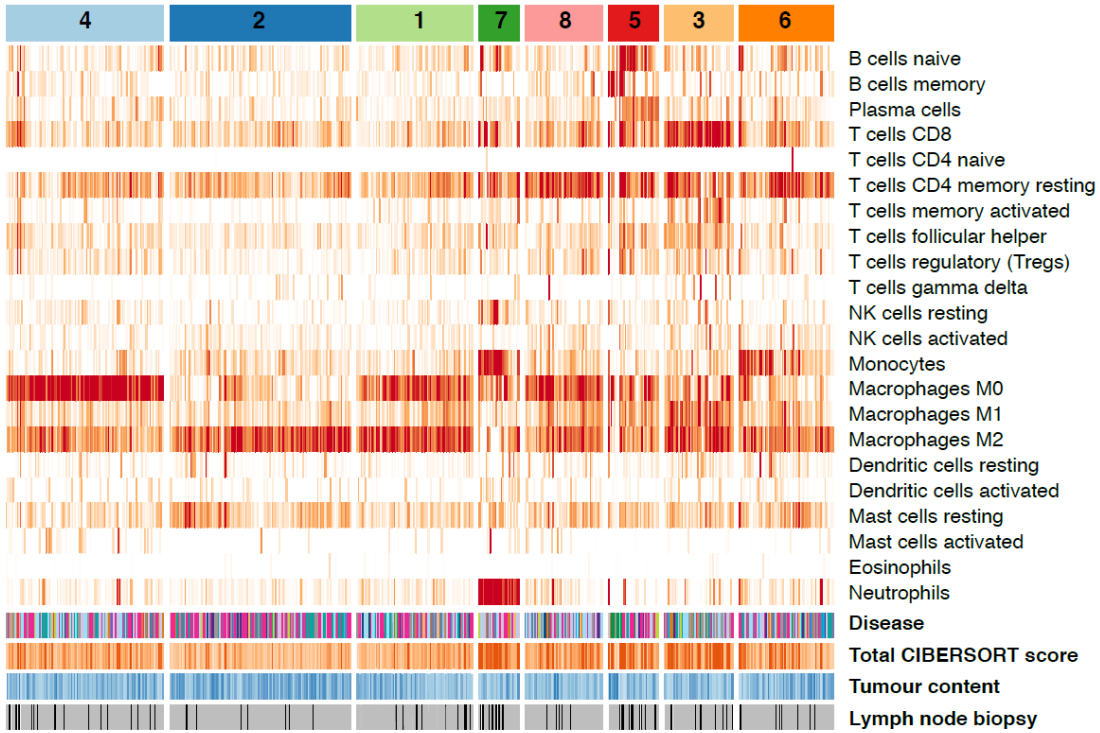
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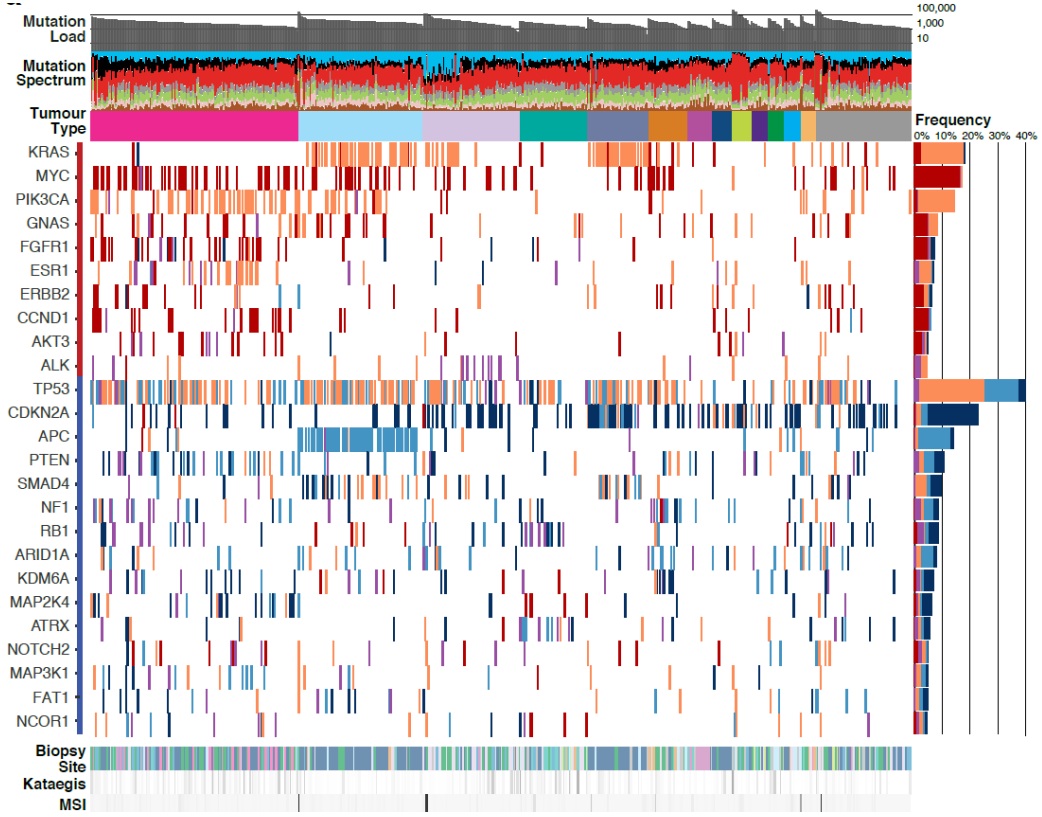


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Heatmaps / oncoplots

- Can use eg. R complexheatmap/pheatmap, maftools

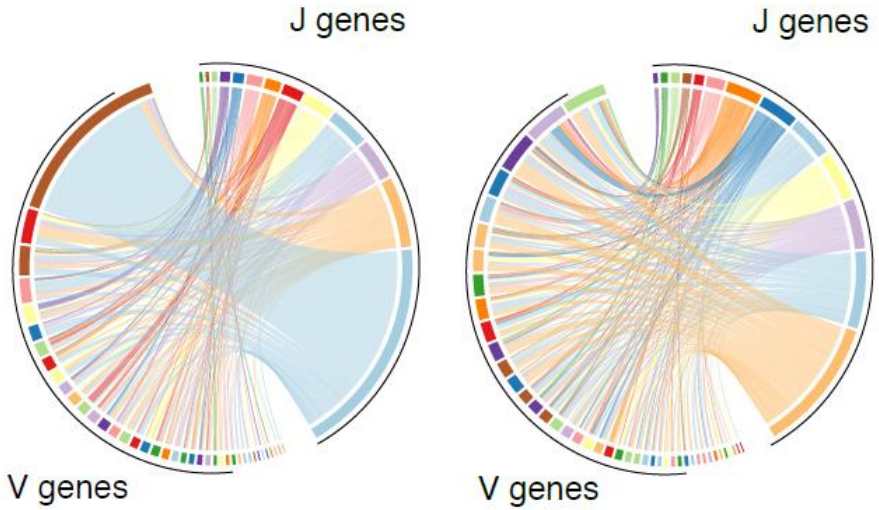




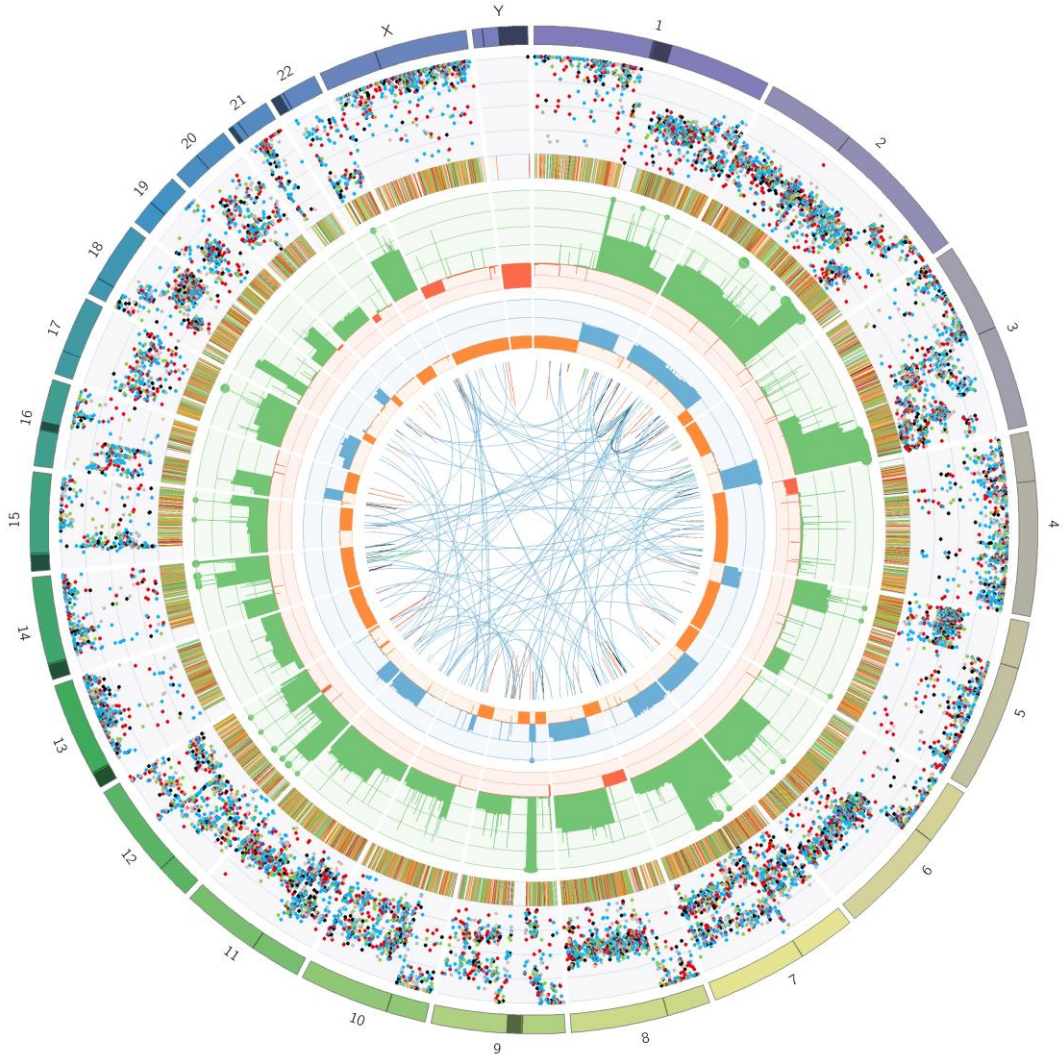
PMID: 35121966

Circos diagrams

- Can use eg. R circlize, Perl circos
  - Check out mk.bcgsc.ca, circos.ca



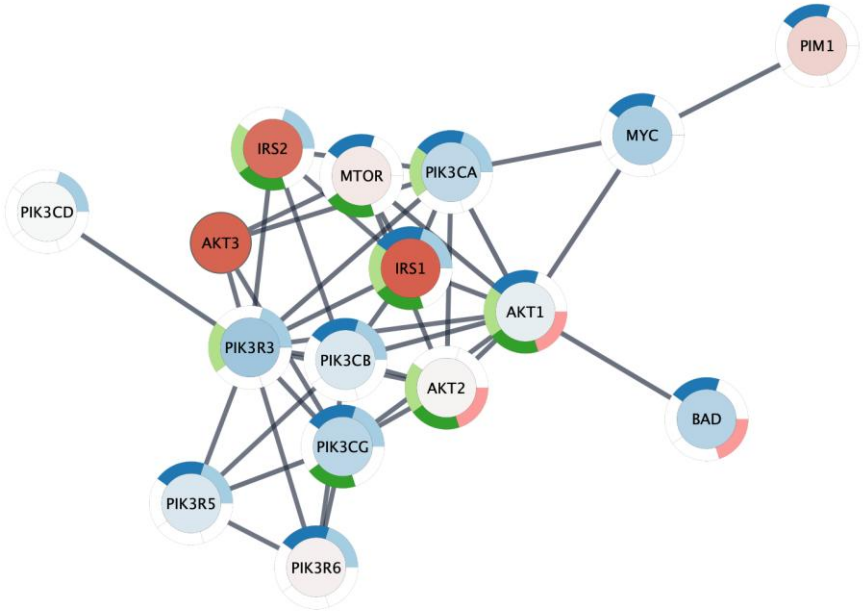
PMID: 35121966



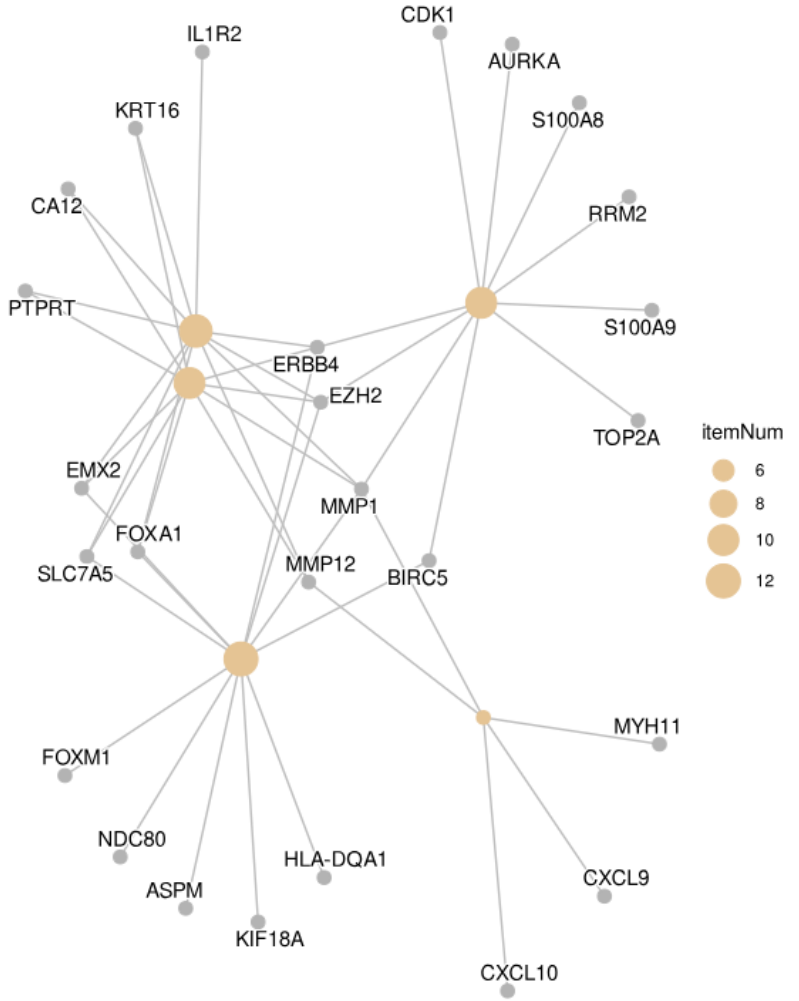
Circos diagram from Purple
(https://github.com/hartwigmedical/hmftools/blob/master/purple)

Pathways/networks

- Can use eg. R clusterprofiler, Cytoscape



https://cytoscape.org/cytoscape-tutorials/protocols/pathway-figure-gene-set-analysis



https://yulab-smu.top/biomedical-knowledge-mining-book/enrichplot-networks.html

Genome browsers

- Sequencing data files are huge
  - Bam/cram file of a human genome is 100-300GB
  - Require specialized tools for fast interactive viewing

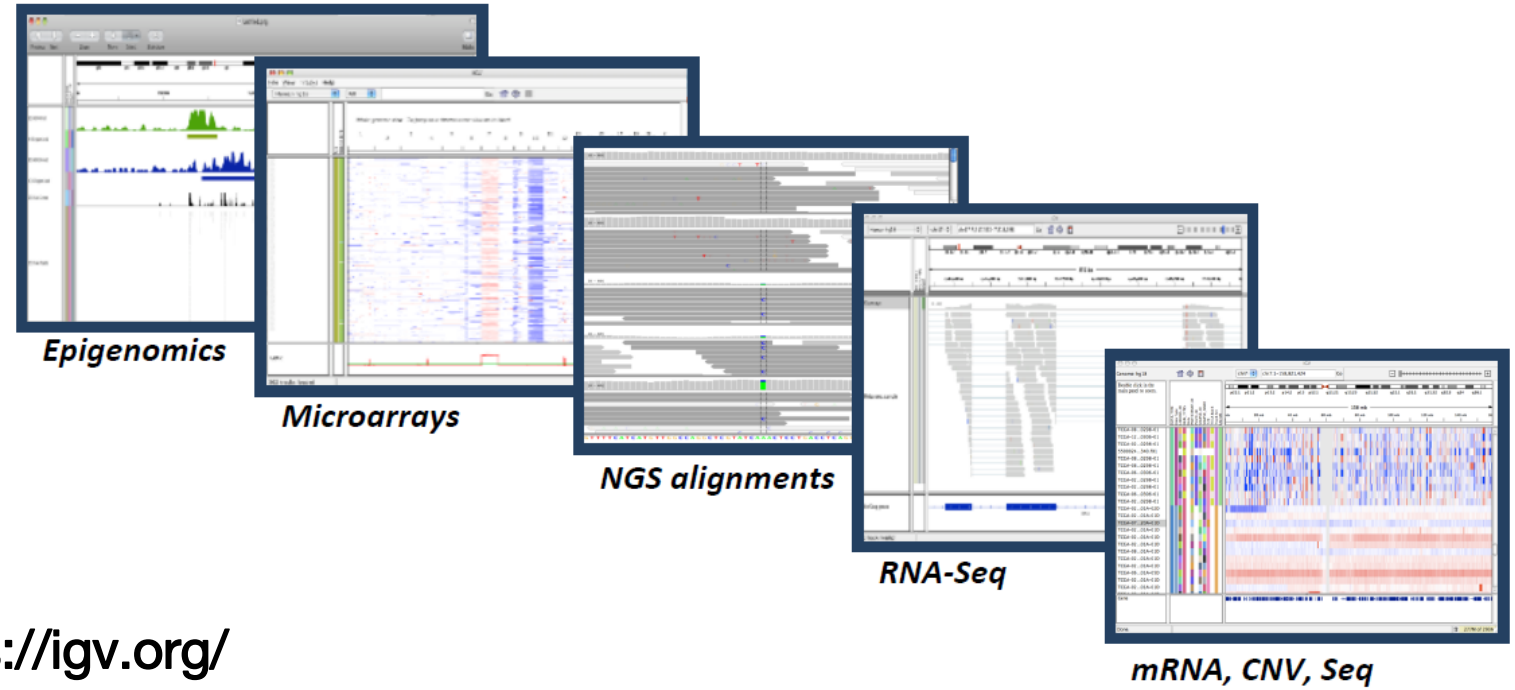
- Typical tasks:
  - Visualizing reads
  - Inspecting variants
  - Viewing with genomic features

Popular genome browsers

	• Runs on local computer • Excellent for bam/cram visualization • The first choice for variant review
	• Runs in browser • Excellent for viewing public genomic data (annotations, regulatory elements, conservation, variation) • Can add custom data tracks
	• Runs in browser • Excellent for comparing transcript models and variant effects • Ensembl annotations frequently used in genomic annotation pipelines
	• Locally hosted, runs in browser • Excellent for visualizing and sharing custom/local genomic datasets

Integrative Genomics Viewer (IGV)

Desktop application for the interactive
visual exploration of integrated genomic datasets



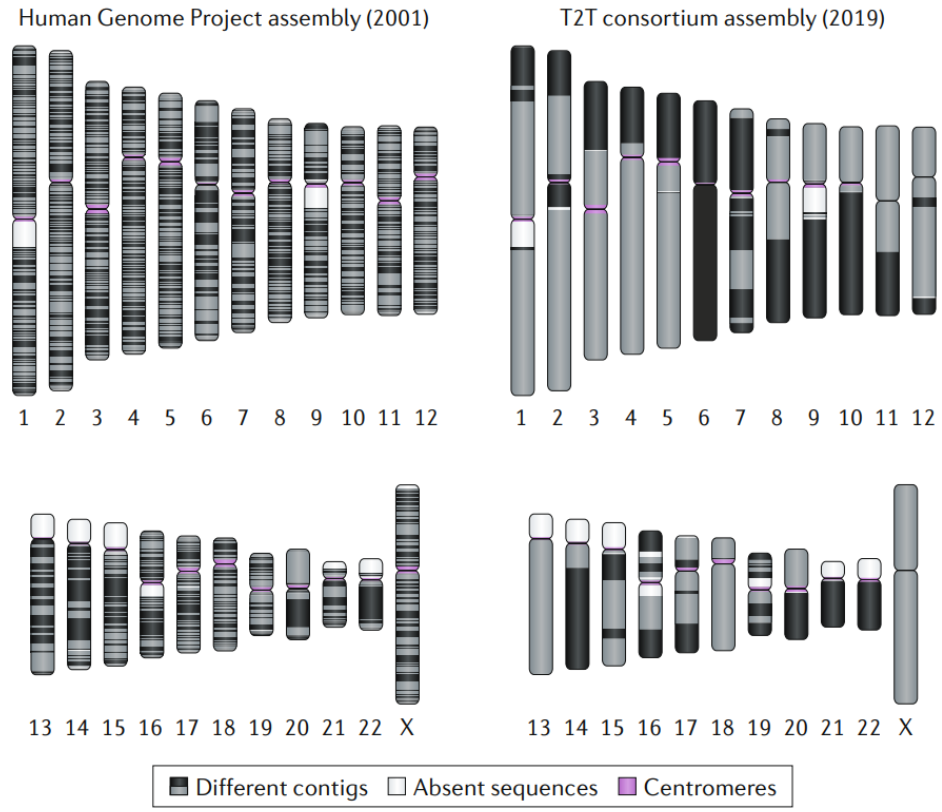
https://igv.org/

Features

- With IGV you can…
  - Explore large genomic datasets with an intuitive, easy-to-use interface
  - Integrate multiple data types with clinical and other sample information
  - View data from multiple sources: local, remote, and “cloud-based”
  - Automation of specific tasks using command-line interface

A note about genome assemblies

- Reads are usually shown in alignment to the **reference genome** for a species
- Reference genomes are imperfect and being improved
- For human:
  - hg19/GRCh37: Older version, still used in clinical settings
  - hg38/GRCh38: Newest version of a the standard reference
    - Many annotations available
    - Most commonly used in genomics
    - Still contains gaps, does not capture all variation
  - T2T-CHM13: Telomere-to-telomere assembly from one haplotype
    - Fewer gaps, excellent for repetitive regions
    - Incomplete annotation resources
  - Pangenome: Graph-based
    - Not just one version of the reference - captures the most variation
    - Requires specialized approaches to make use of it



PMID: 32504078

IGV data sources

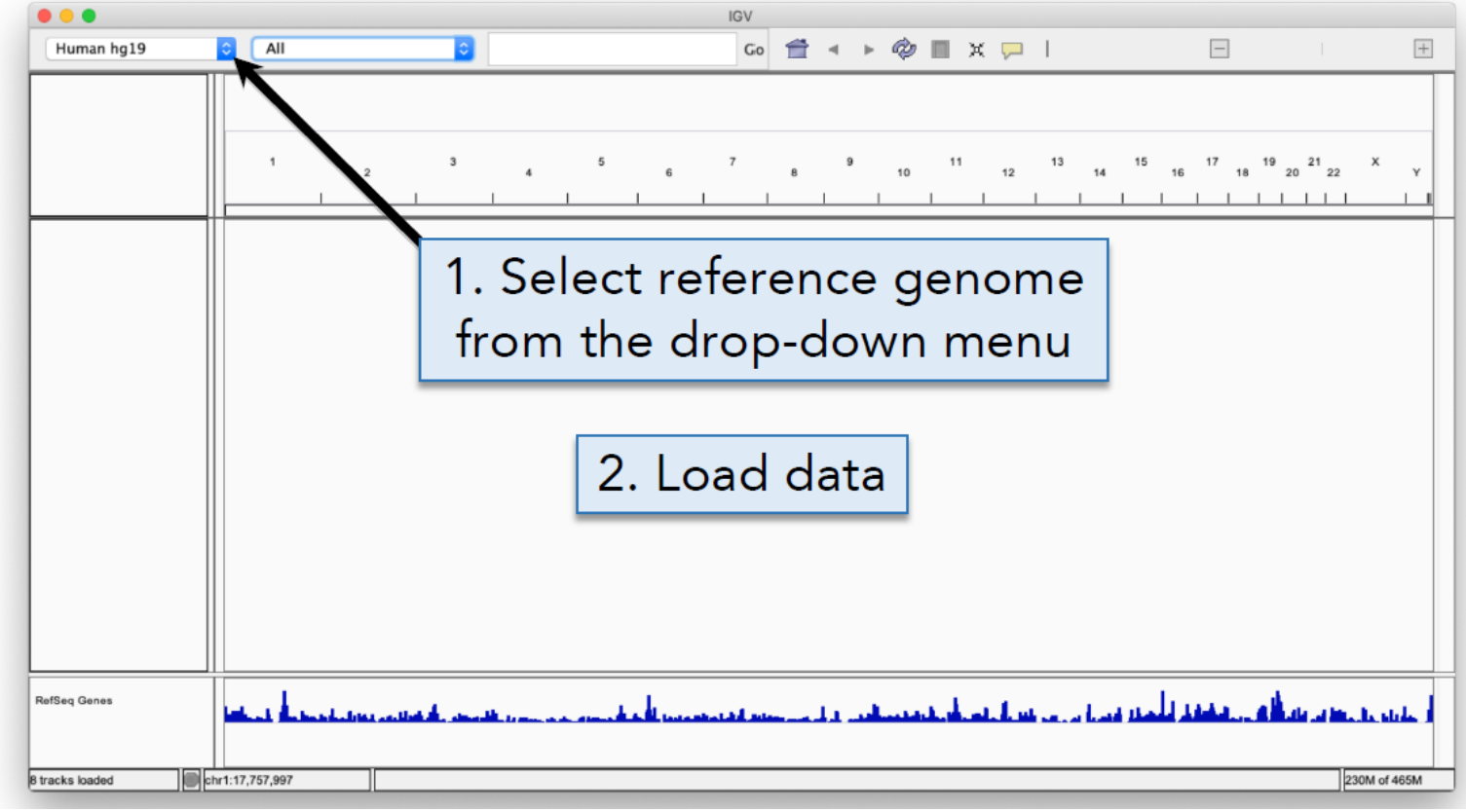


- View **local files** without uploading
- View **remote files** with downloading the whole dataset

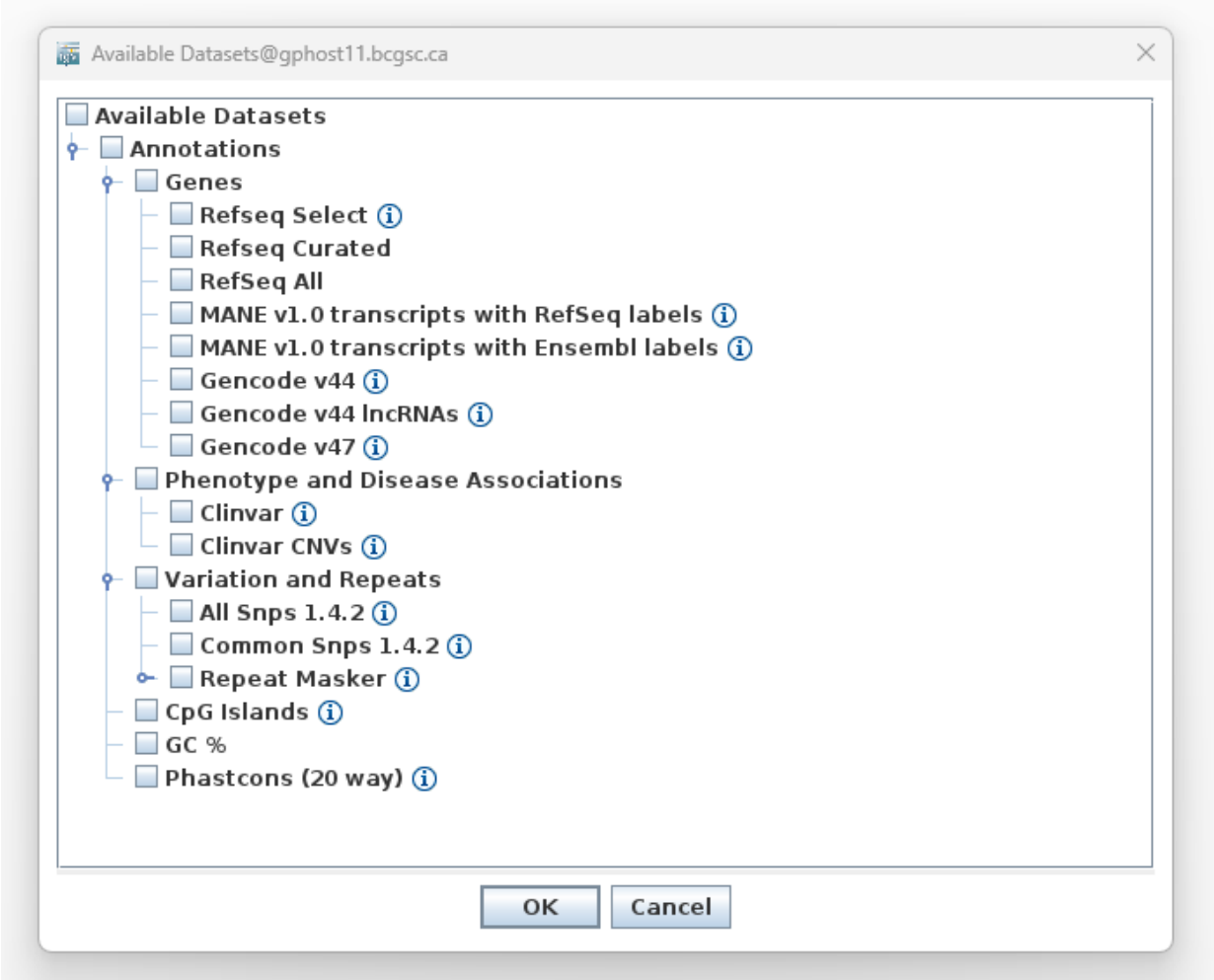
Using IGV: The basics

- Launch IGV
- Select a reference genome
- Load data
- Navigate through the data
  - WGS data
  - SNVs
  - Structural variants

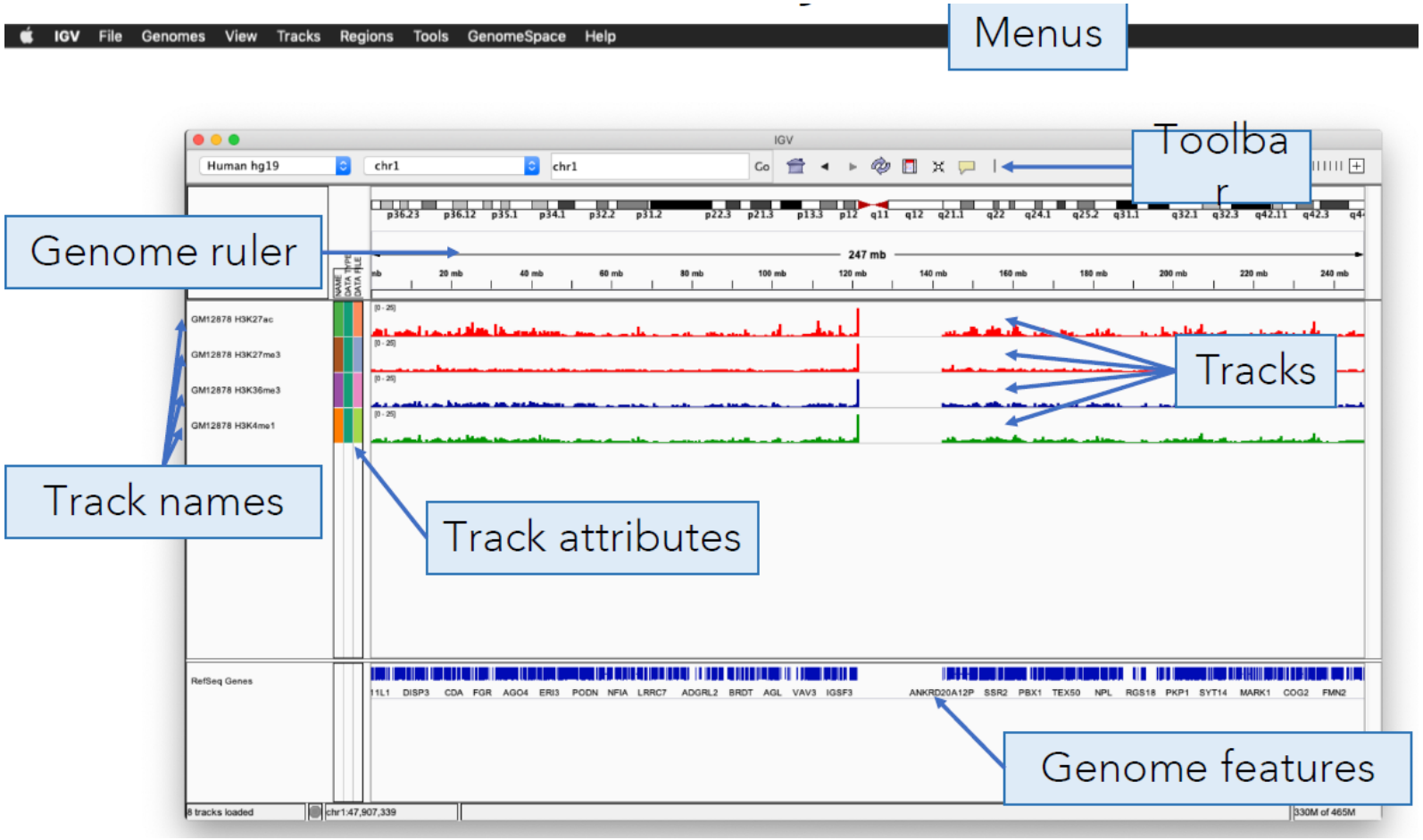
Launch IGV and select genome



Load data


Available Datasets@gphost11.bcgsc.ca
Available Datasets
Annotations
Genes
Refseq Select
Refseq Curated
RefSeq All
MANE v1.0 transcripts with RefSeq labels
MANE v1.0 transcripts with Ensembl labels
Gencode v44
Gencode v44 lncRNAs
Gencode v47
Phenotype and Disease Associations
Clinvar
Clinvar CNVs
Variation and Repeats
All Snps 1.4.2
Common Snps 1.4.2
Repeat Masker
CpG Islands
GC %
Phastcons (20 way)
OK
Cancel

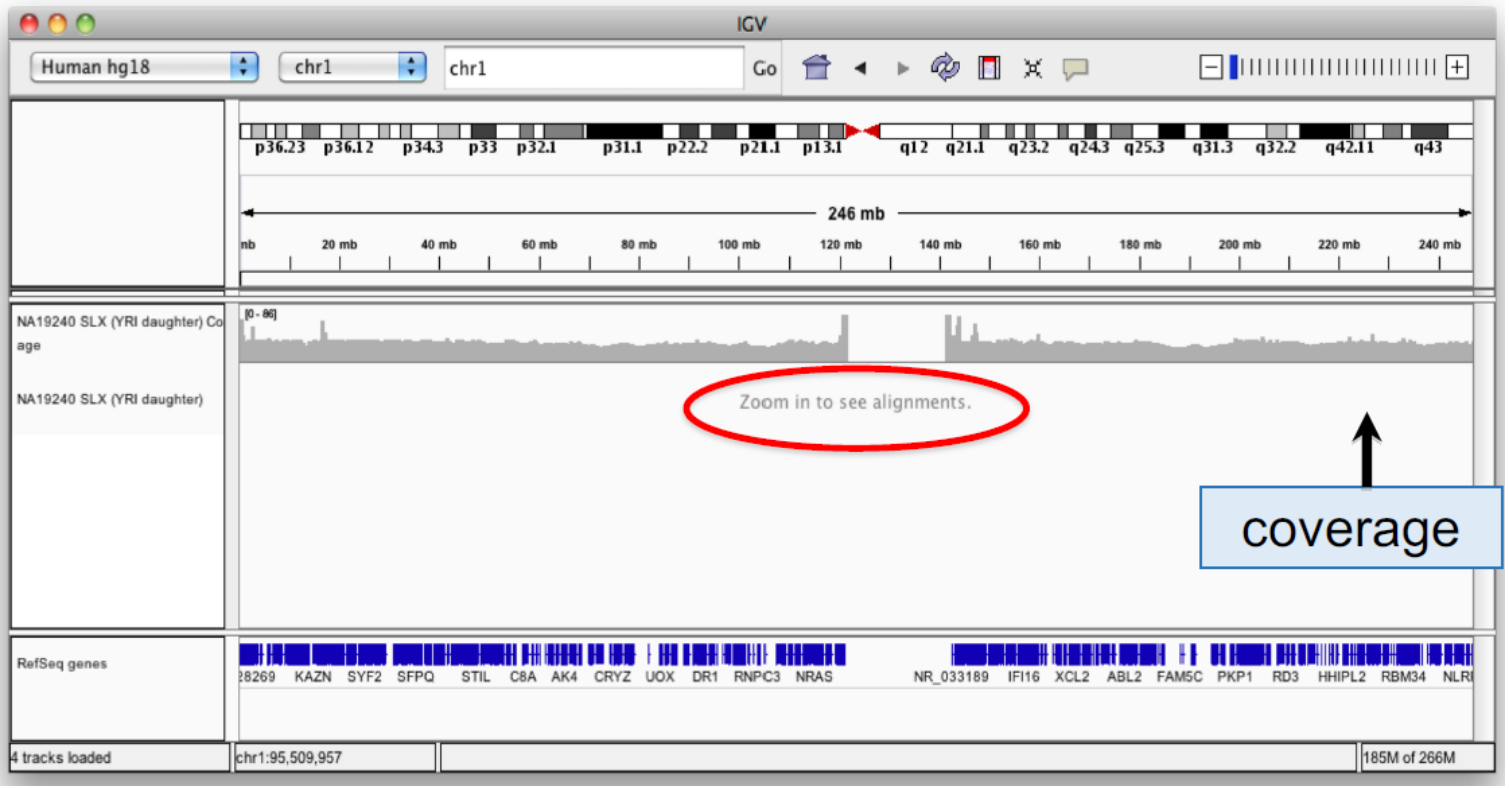
Screen layout



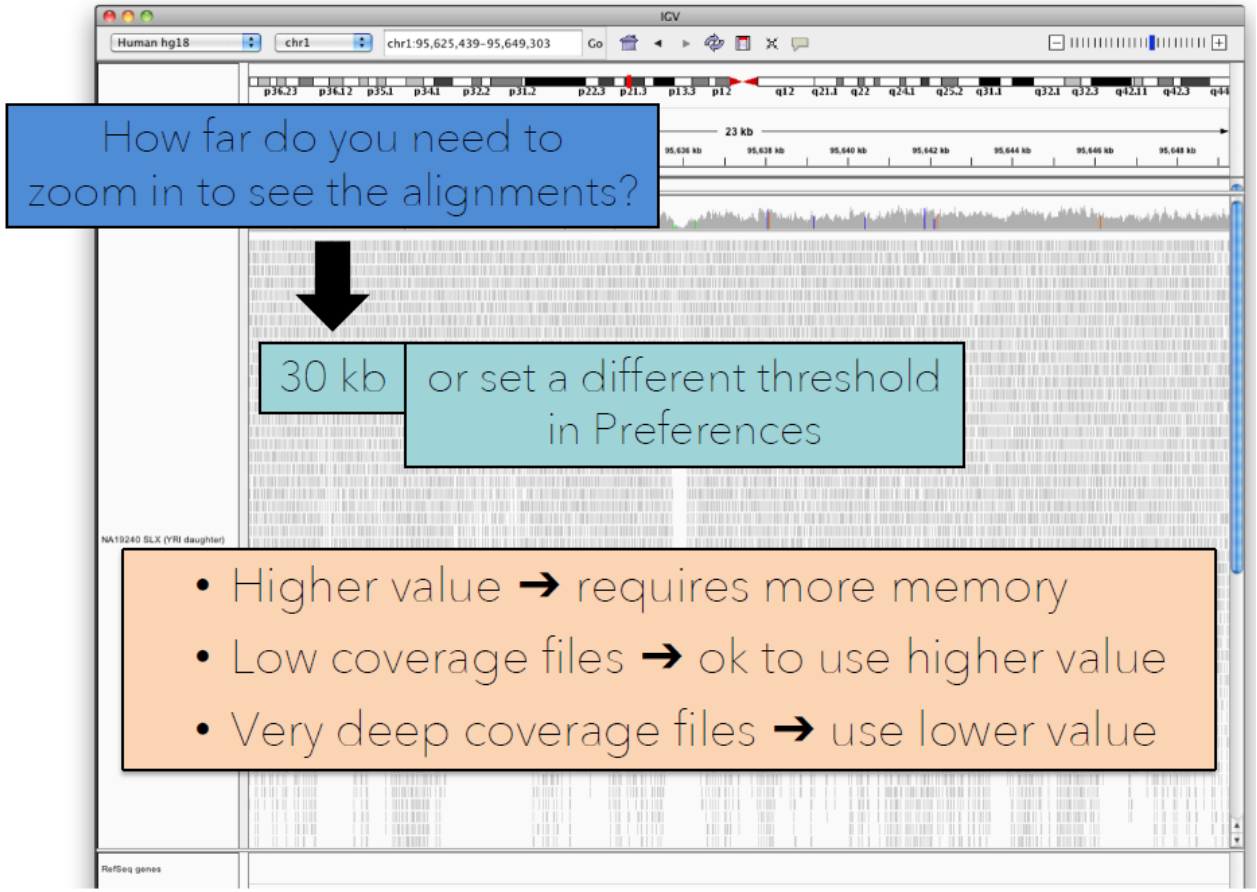
File formats and track types

- The file format defines the track type
- The track type determines the display options
- Compatible data formats:
  - BAM
  - BED
  - BedGraph
  - bigBed
  - bigWig
  - Birdsuite Files
  - broadPeak
  - CBS
  - Chemical Reactivity Probing Profiles
  - chrom.sizes
  - CN
  - Custom File Formats
  - Cytoband
  - FASTA
  - GCT
  - CRAM
  - genePred
  - GFF/GTF
  - GISTIC
  - Goby
  - GWAS
  - IGV
  - LOH
  - MAF (Multiple Alignment Format)
  - MAF (Mutation Annotation Format)
  - Merged BAM File
  - MUT
  - narrowPeak
  - PSL
  - RES
  - RNA Secondary Structure Formats
  - SAM
  - Sample Info (Attributes) file
  - SEG
  - SNP
  - TAB
  - TDF
  - Track Line
  - Type Line
  - VCF
  - WIG

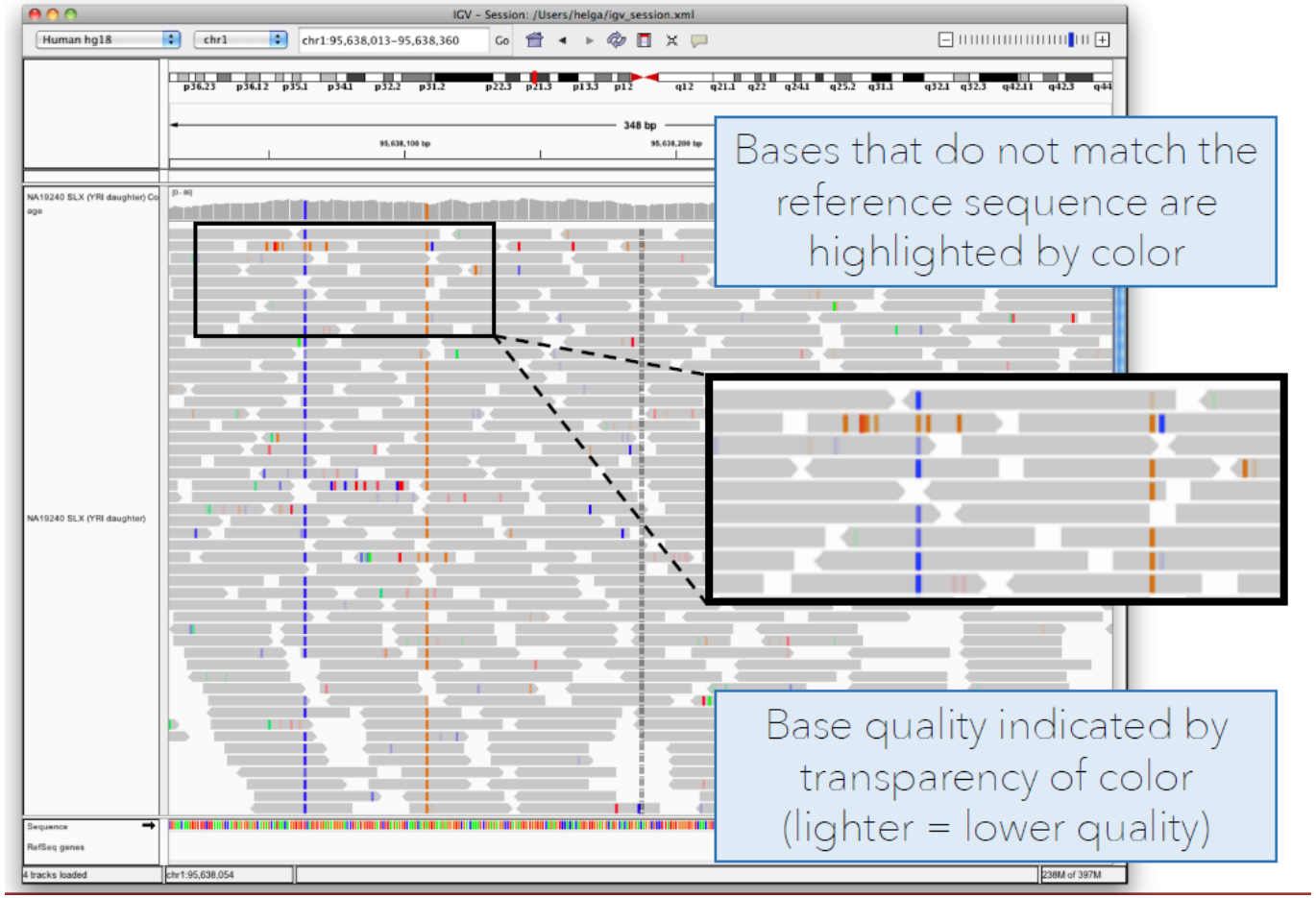
Viewing alignments



Viewing alignments – Zoom in



Viewing alignments – Zoom in

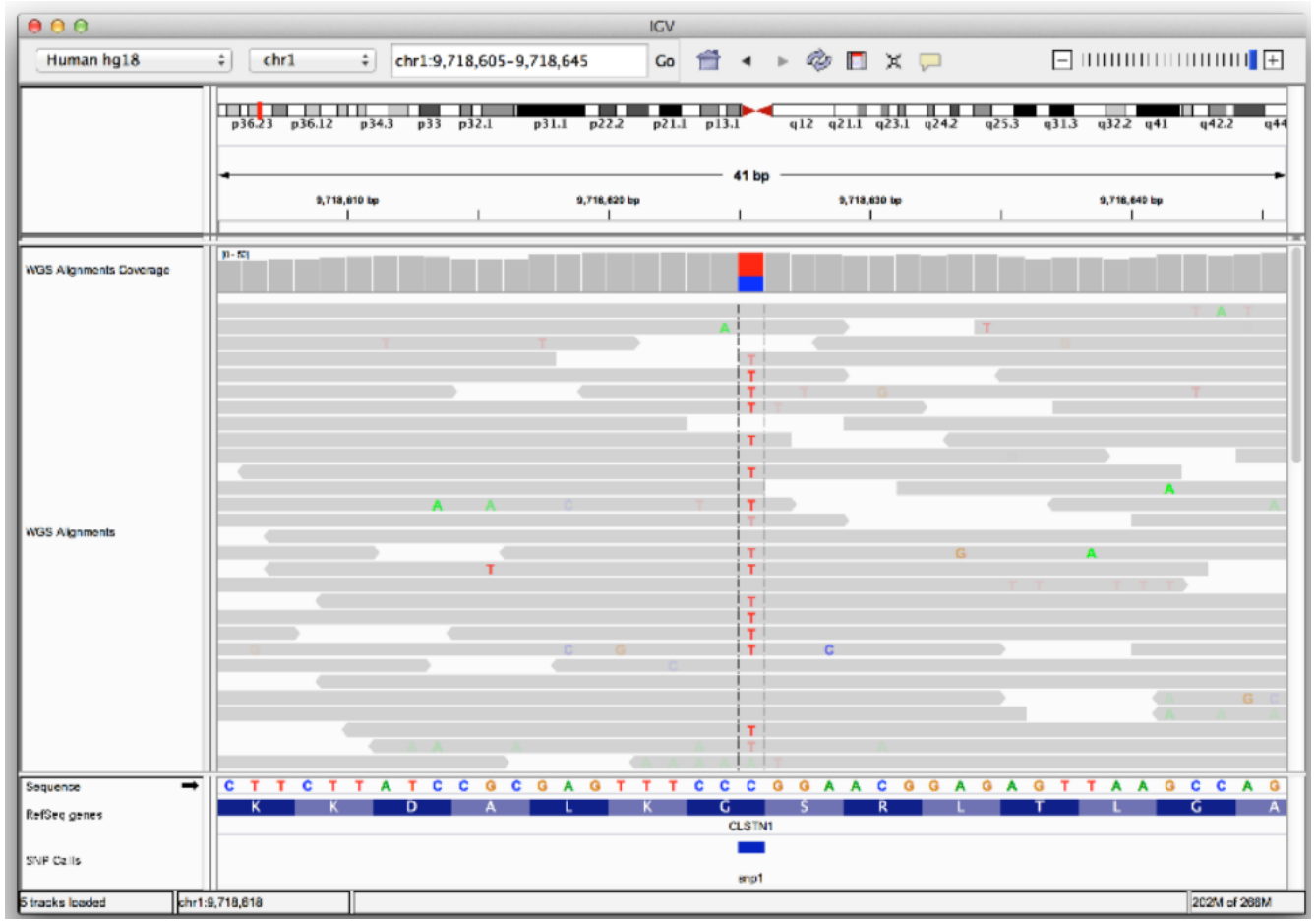


SNVs and structural variants

- Important metrics for evaluating the validity of single nucleotide variants (SNVs):
  - Coverage
  - Amount of support
  - Strand bias / PCR artifacts
  - Mapping qualities
  - Base qualities

- Important metrics for evaluating SVs:
  - Coverage
  - Insert size
  - Read pair orientation

Viewing SNPs and SNVs



- Things to watch for:
  - variant allele frequency (VAF)
  - number of supporting reads
  - poor quality bases
  - nearby mismatches

Viewing SNPs and SNVs



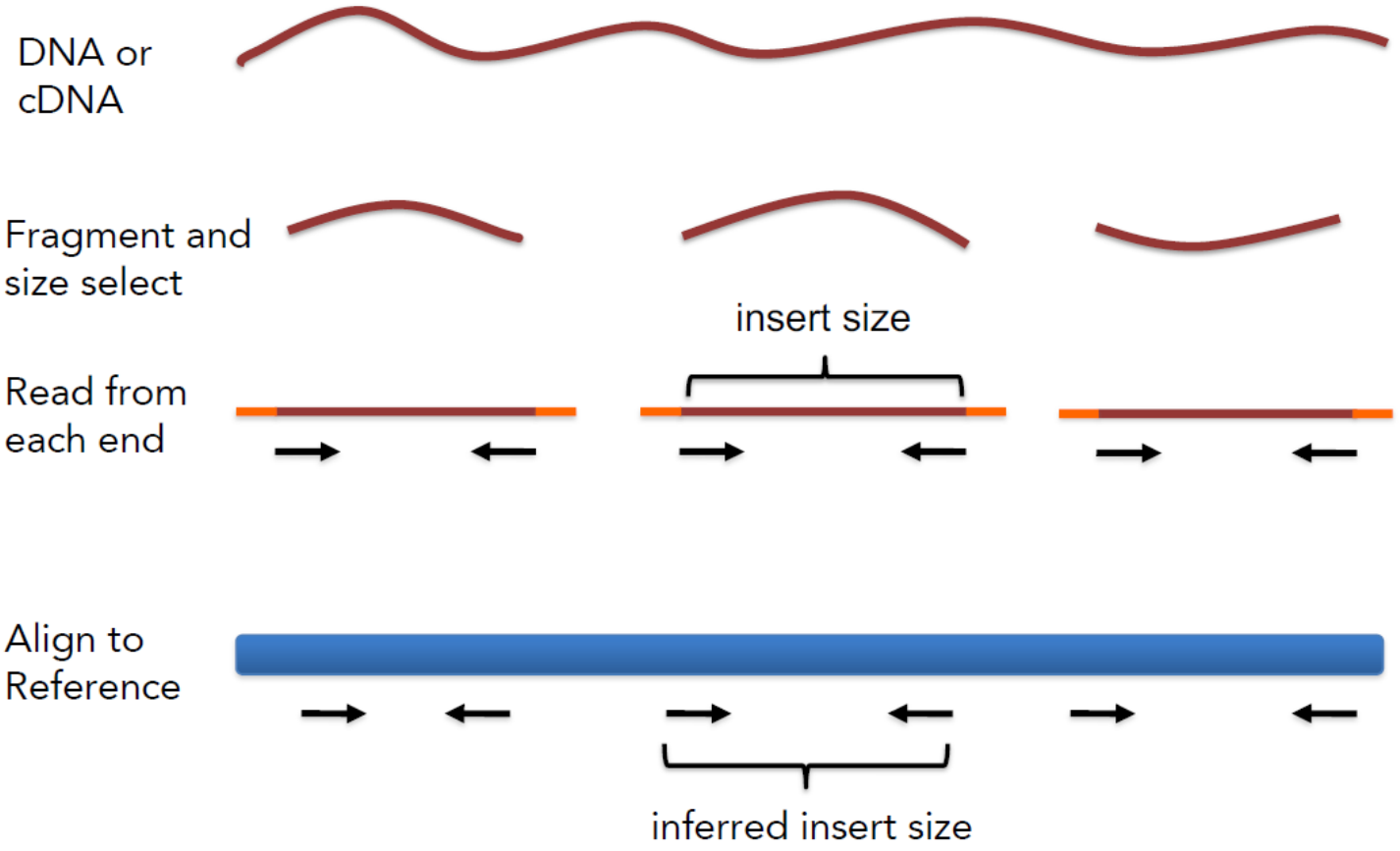
- Things to watch for:
  - strand bias
  - variants near read ends
  - read mapping quality
  - repetitive sequence

Viewing structural variants

- Paired reads can yield evidence for genomic “structural events”, such as deletions, translocations, and inversions

- Alignment coloring options help highlight these events based on:
  - Inferred insert size (template length)
  - Pair orientation (relative strand of pair)

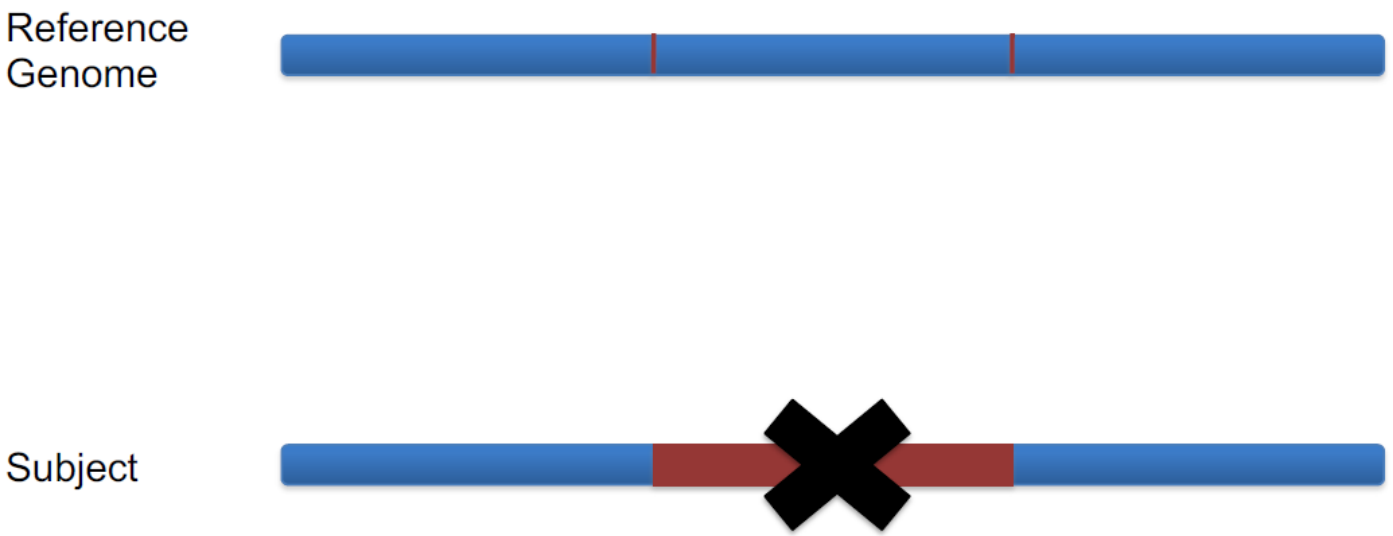
Paired-end sequencing



Interpreting inferred insert size

- The “inferred insert size” can be used to detect structural variants including:
  - **Deletions**
  - **Insertions**

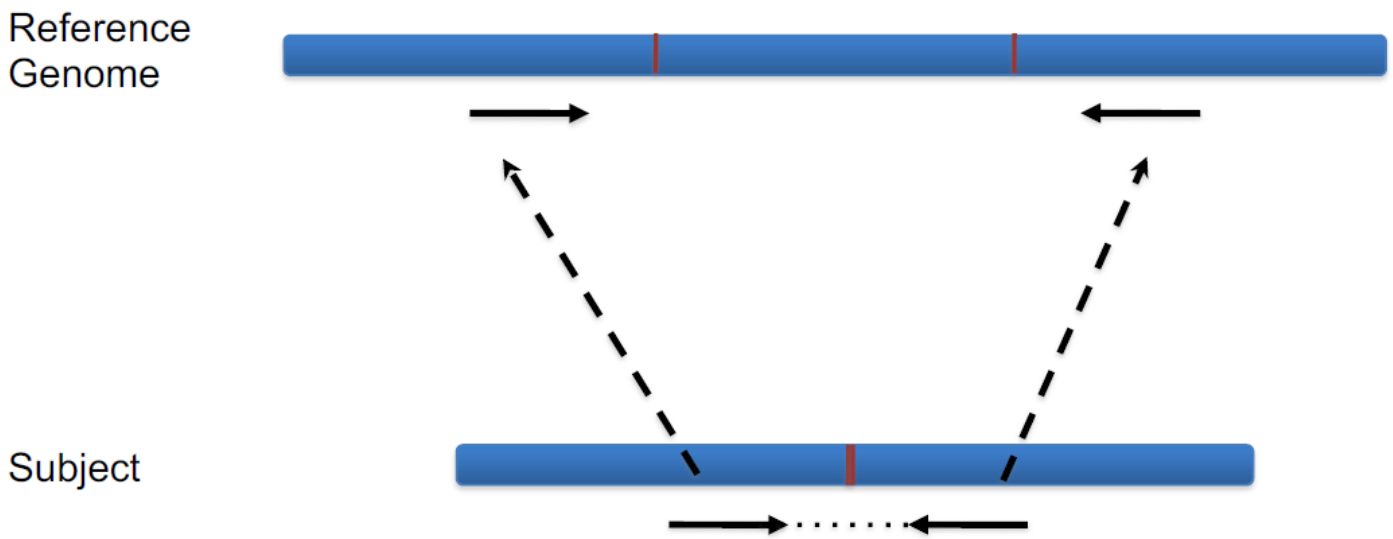
Deletion



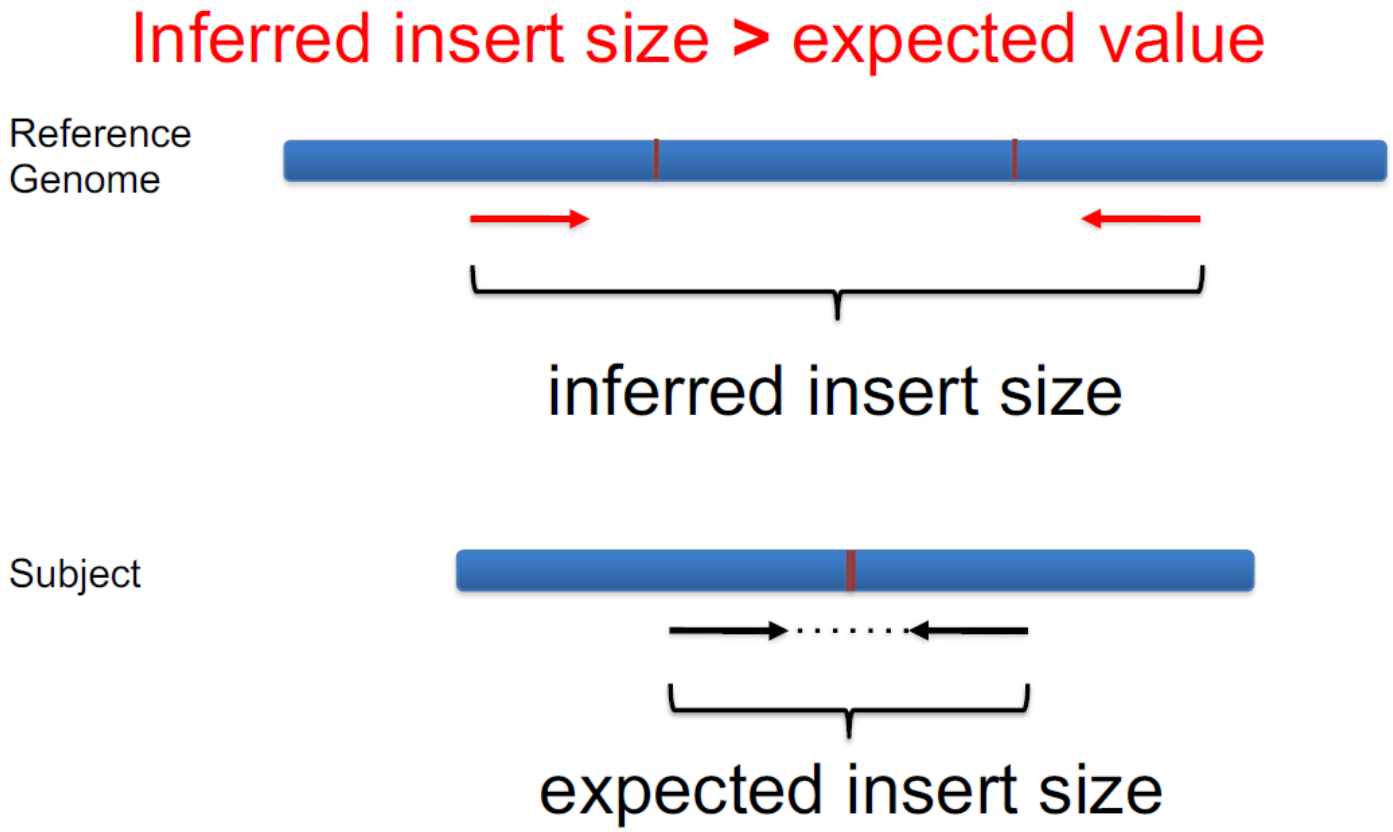
Deletion



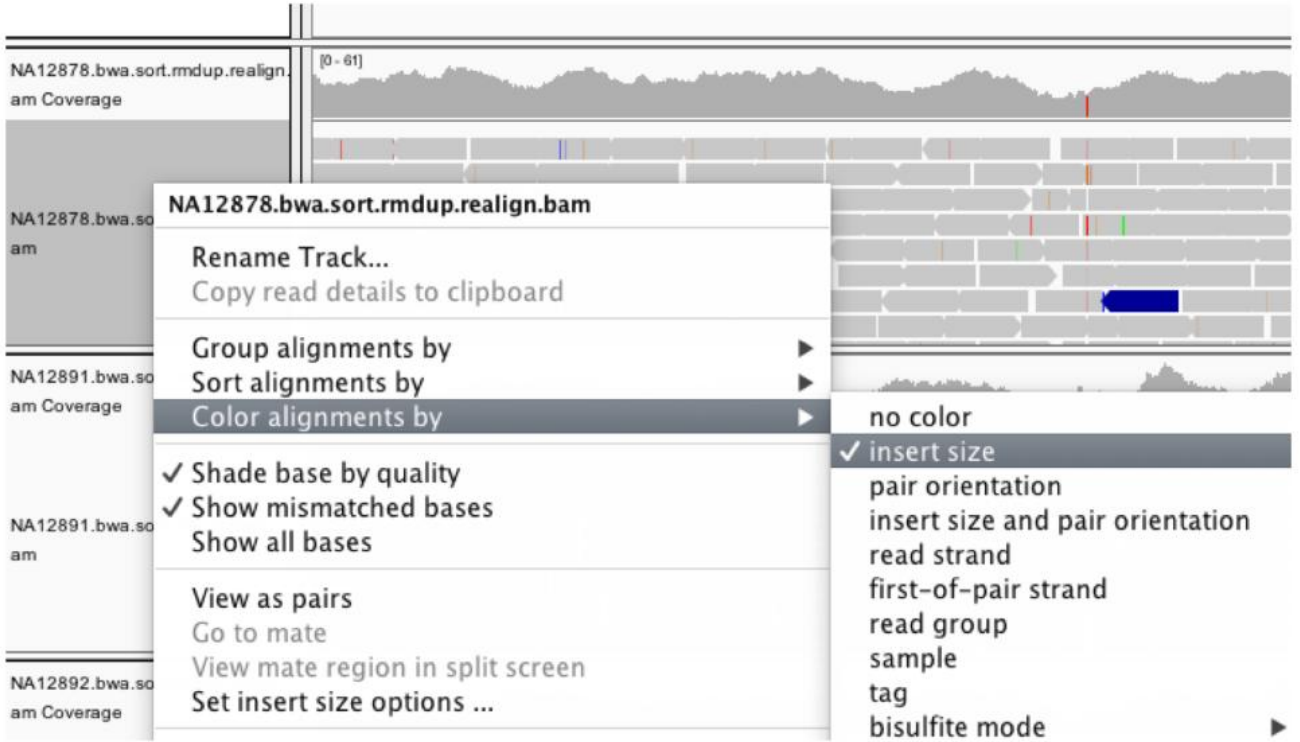
Deletion



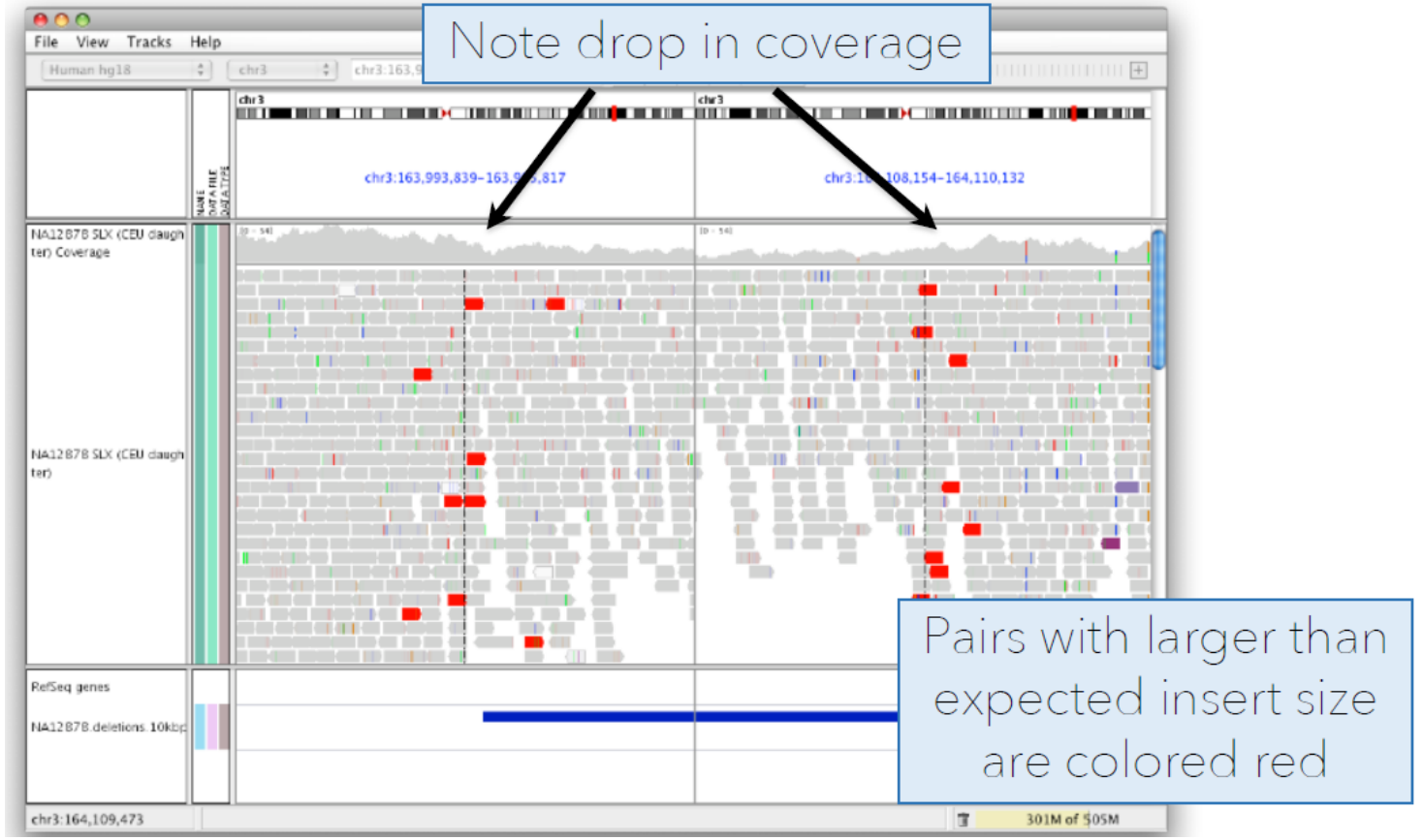
Deletion


Inferred insert size > expected value
Reference Genome
inferred insert size
Subject
expected insert size

Colour by insert size



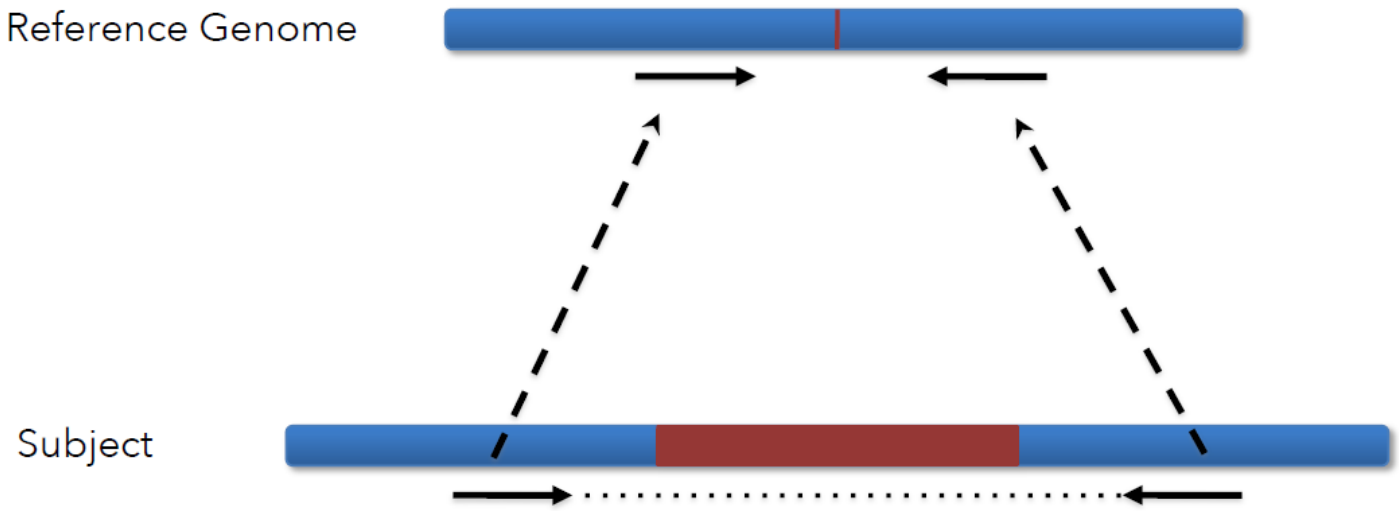
Deletion


Note drop in coverage
File View Tracks Help
chr3:163,993,839-163,
chr3:164,108,154-164,110,132
NA12878 SLX (CEU daugh ter) Coverage
NA12878 SLX (CEU daugh ter)
RefSeq genes
NA12878.deletions.10kbp
chr3:164,109,473
301M of 305M
Pairs with larger than expected insert size are colored red

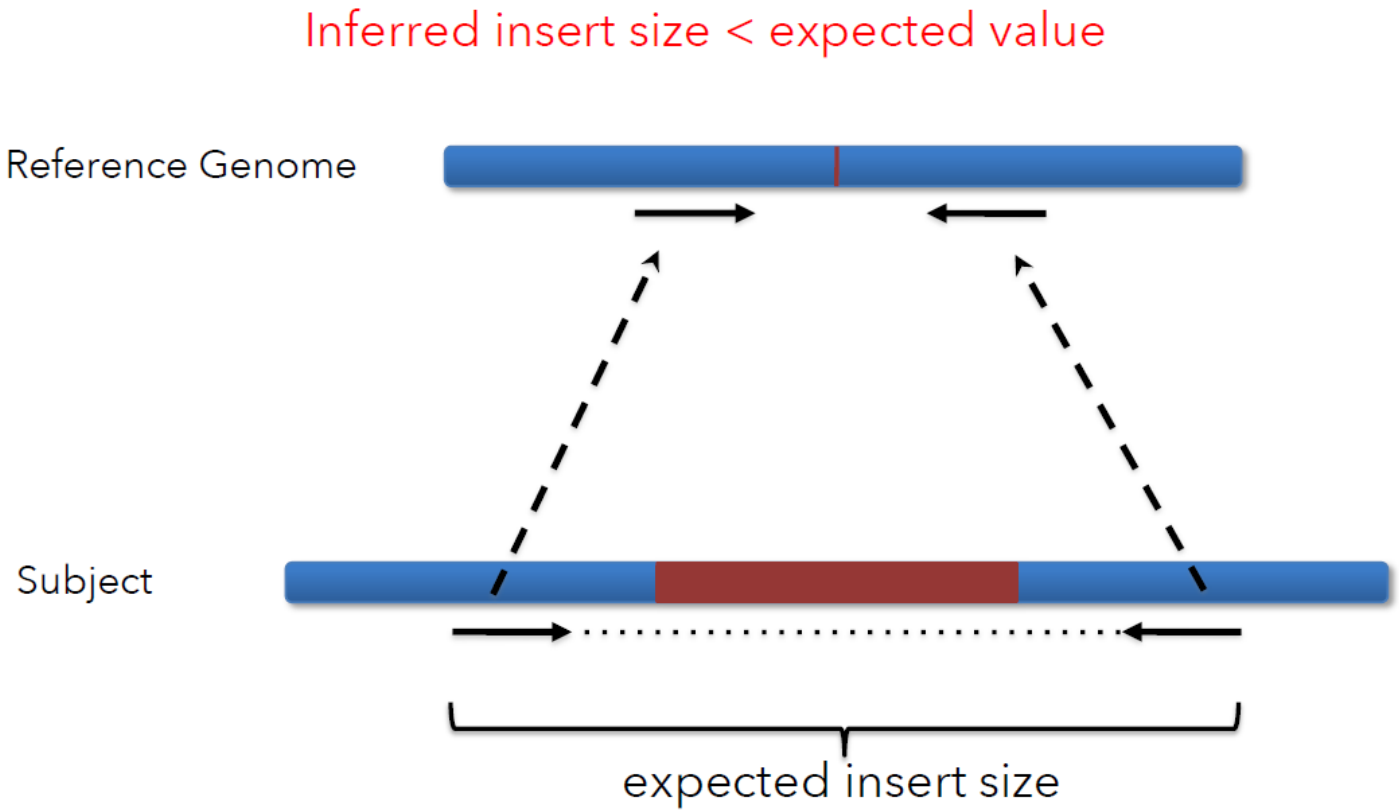
Insertion



Insertion



Insertion

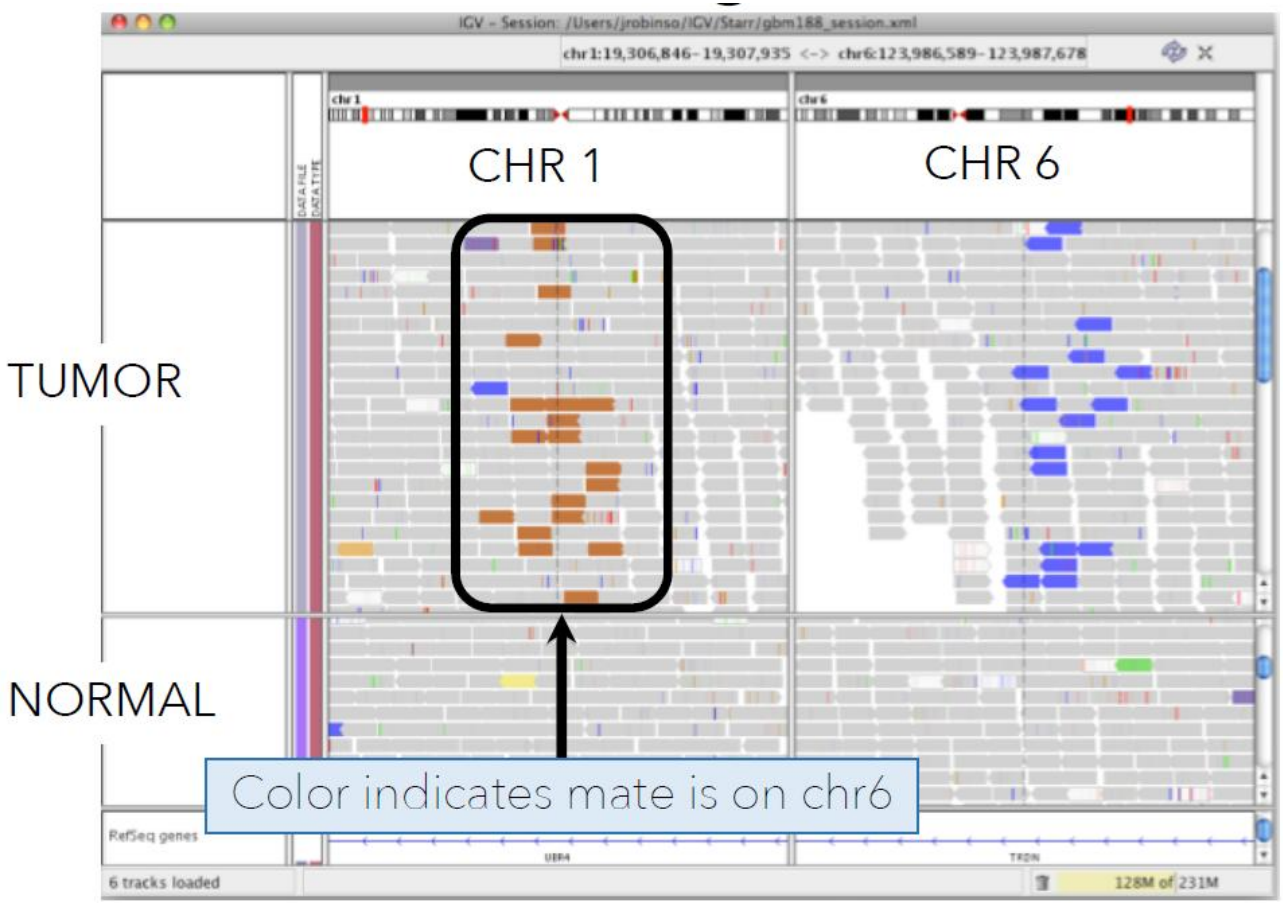

Inferred insert size < expected value
Reference Genome
Subject
expected insert size

Insert size colour scheme

- Smaller than expected insert size:
- Larger than expected insert size:
- Pairs on different chromosomes: each end colored by chromosome of its mate:



Translocation

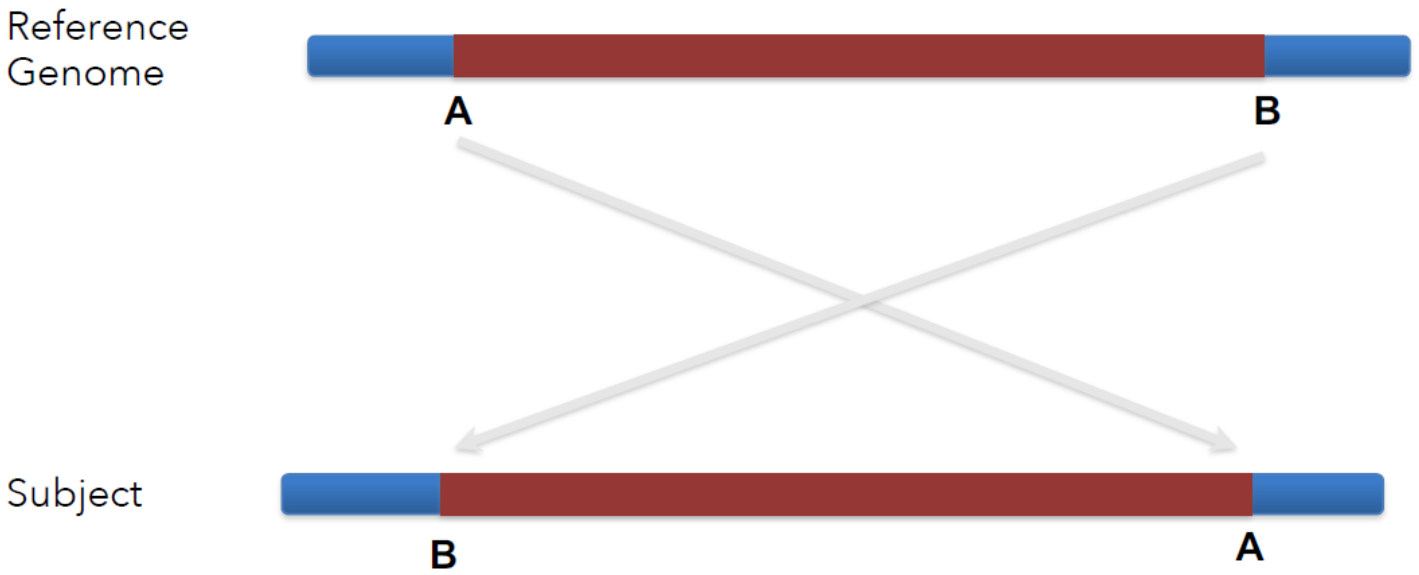


Interpreting read-pair orientations

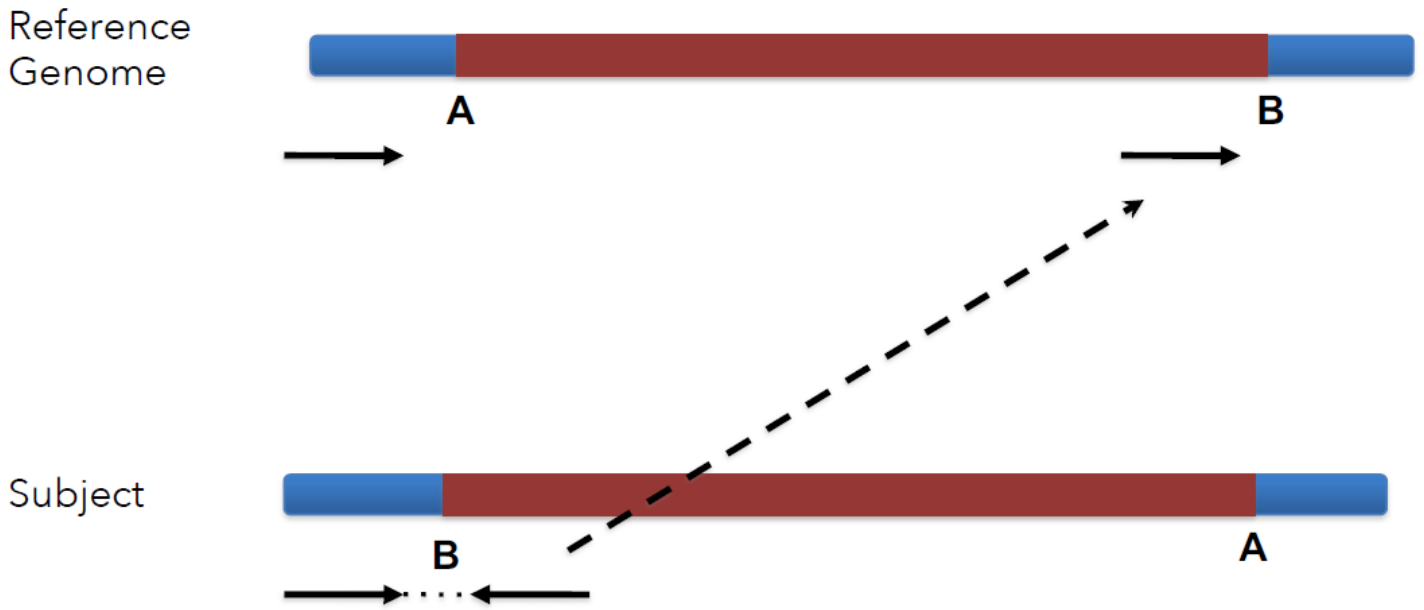
- Orientation of paired reads can reveal structural events:
  - Inversions
  - Duplications
  - Translocations
  - Complex rearrangements

- Orientation is defined in terms of
  - read strand, left vs right, and
  - read order, first vs second

Inversion



Inversion

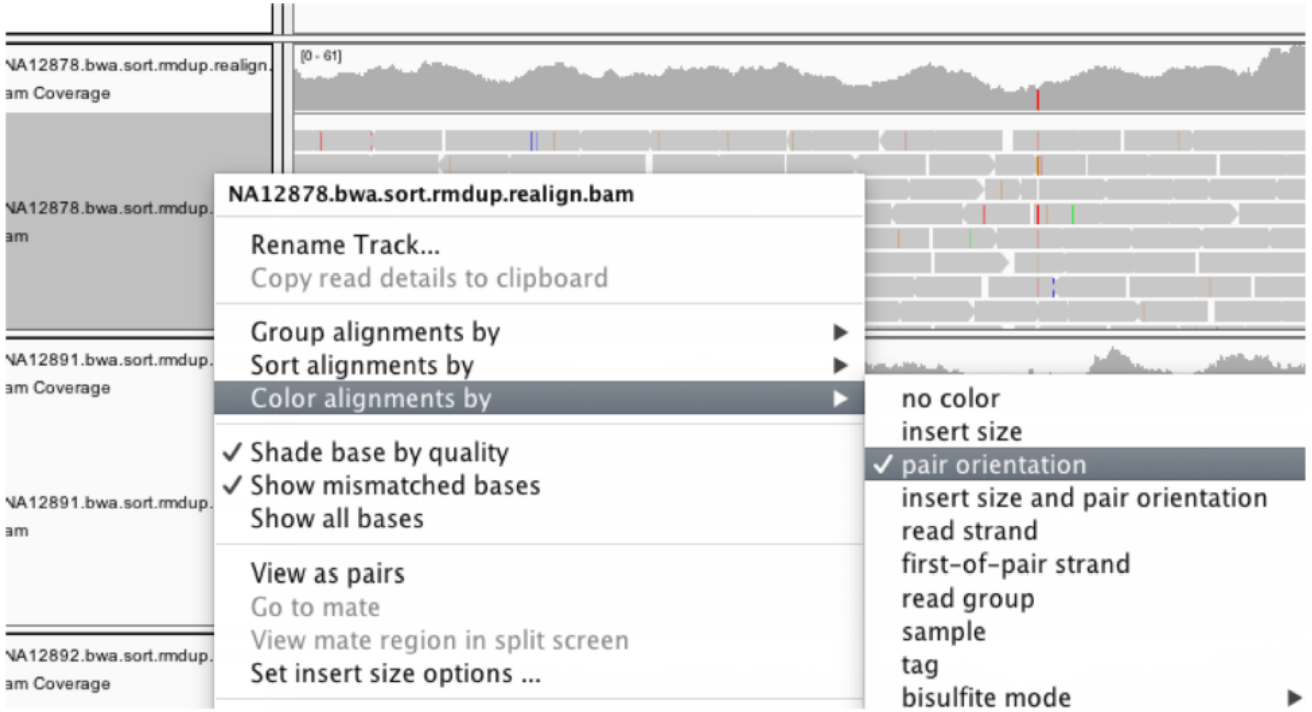


Inversion

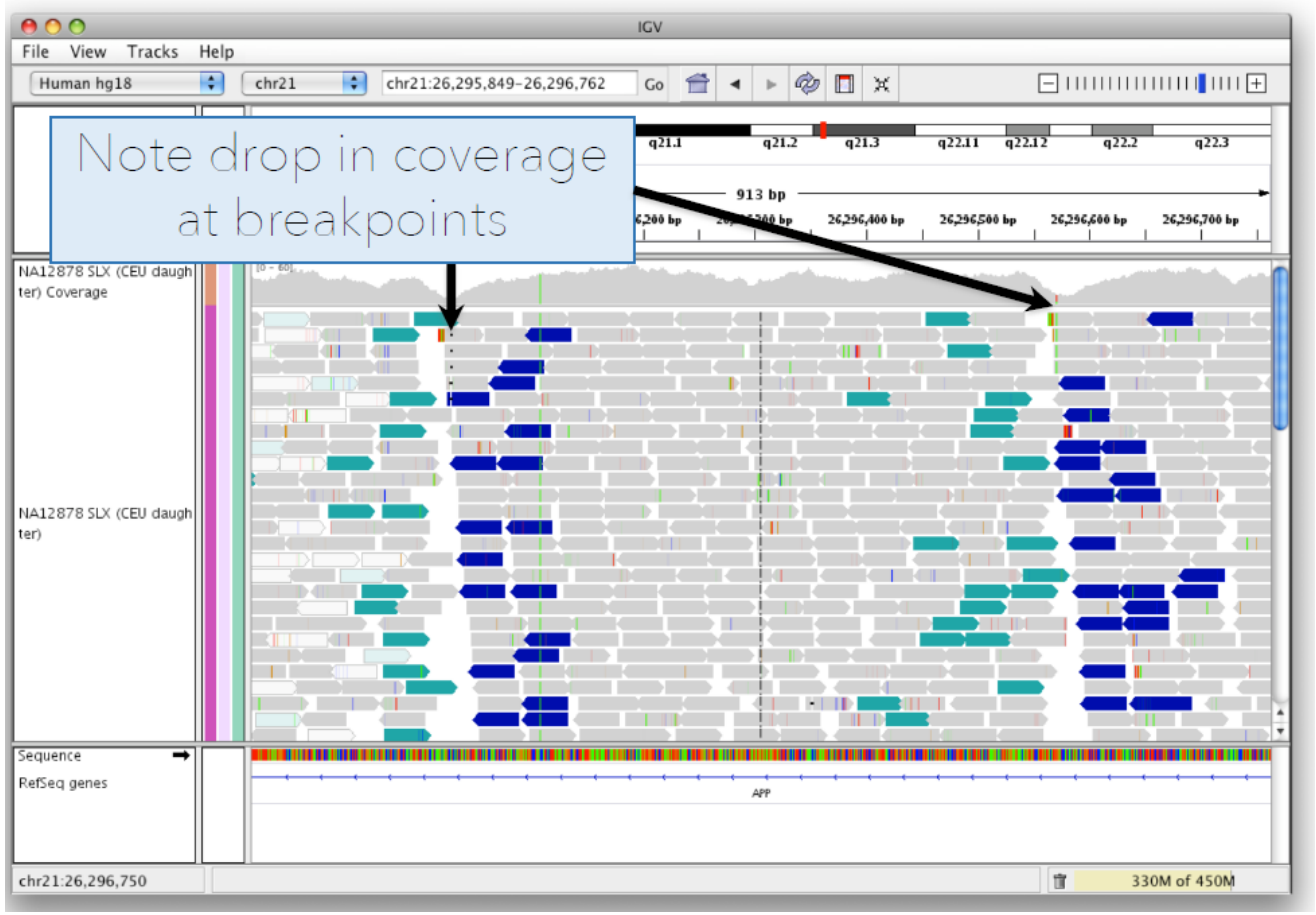


- Change in read pair **orientation and insert size**

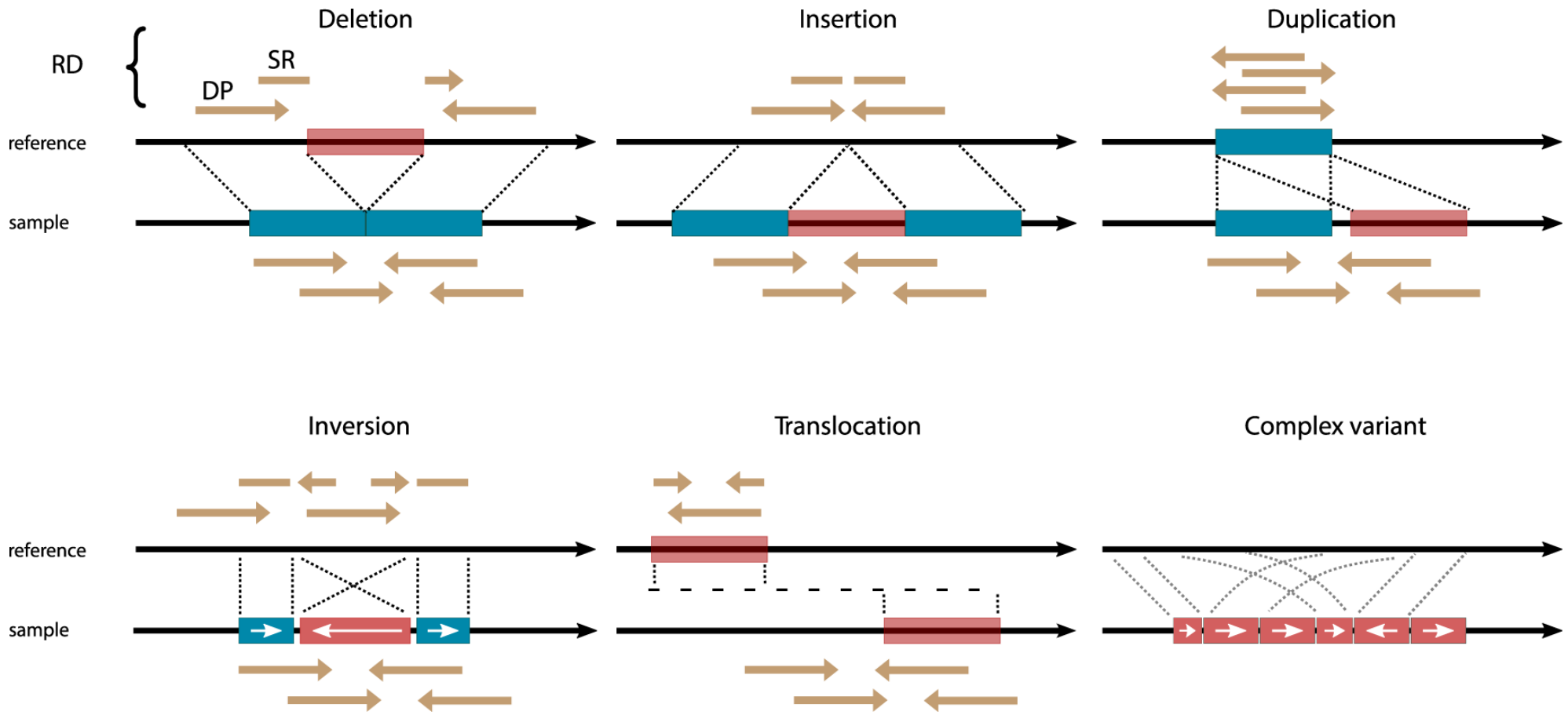
Colour by pair orientation



Inversion


IGV
File View Tracks Help
Human hg18
chr21
chr21:26,295,849-26,296,762
Go
Note drop in coverage at breakpoints
913 bp
NA12878 SLX (CEU daughter) Coverage
NA12878 SLX (CEU daughter)
Sequence
RefSeq genes
APP
chr21:26,296,750
330M of 450M

Summary of SV interpretation



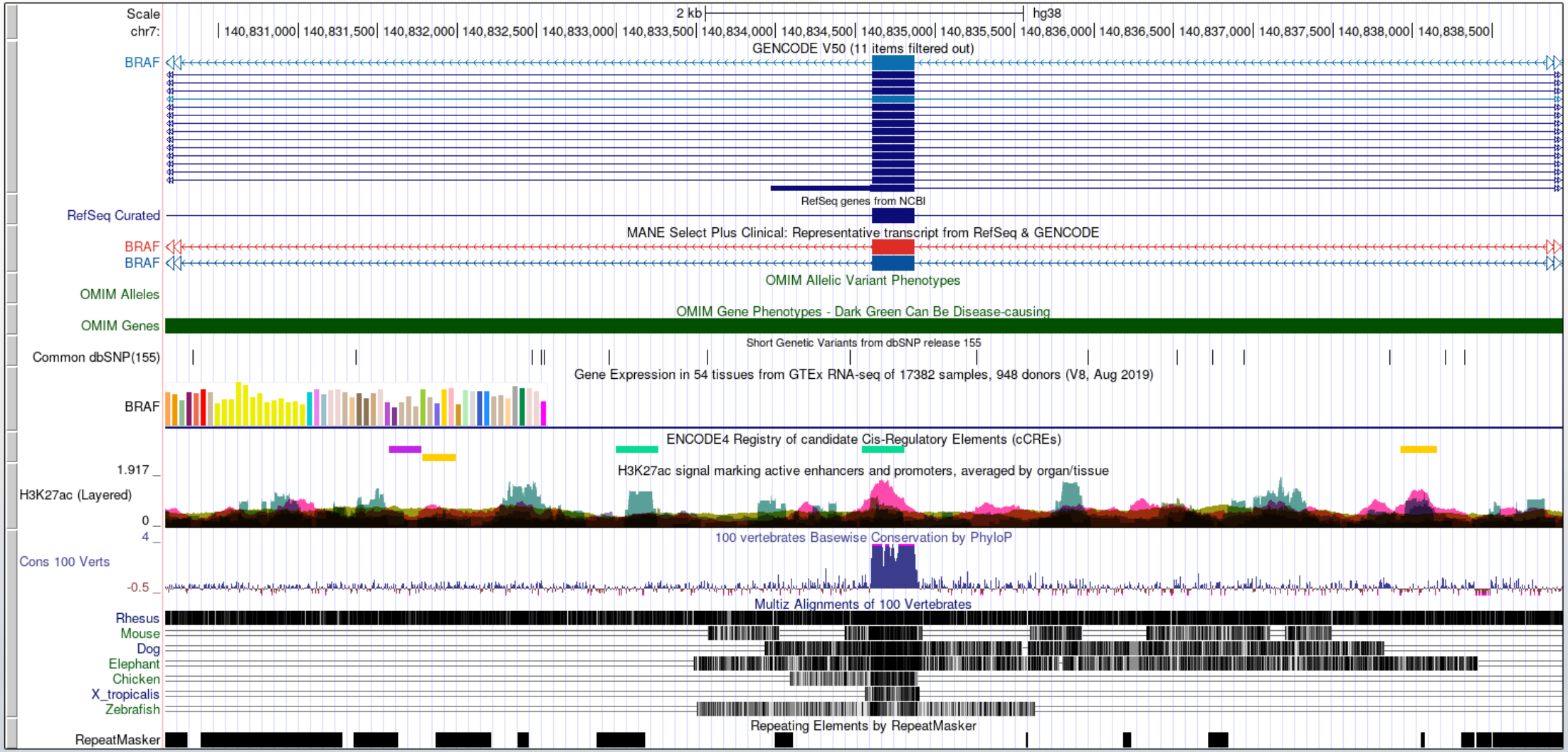
PMID: 33654267

Online genome browsers: UCSC

- **Rich annotations, easy web access**
- Less suitable for detailed review of BAM/CRAM sequencing alignments

- View genes, transcripts, variants, regulatory regions, conservation, and many other annotations
- Navigate from whole chromosomes down to individual nucleotides
- Integrates hundreds of public datasets (ClinVar, dbSNP, ENCODE, GTEx, etc.)
- Supports custom tracks for visualizing your own BED, VCF, BigWig, and other data files
- Includes tools for sequence retrieval, coordinate conversion (liftOver), and data mining (Table Browser)

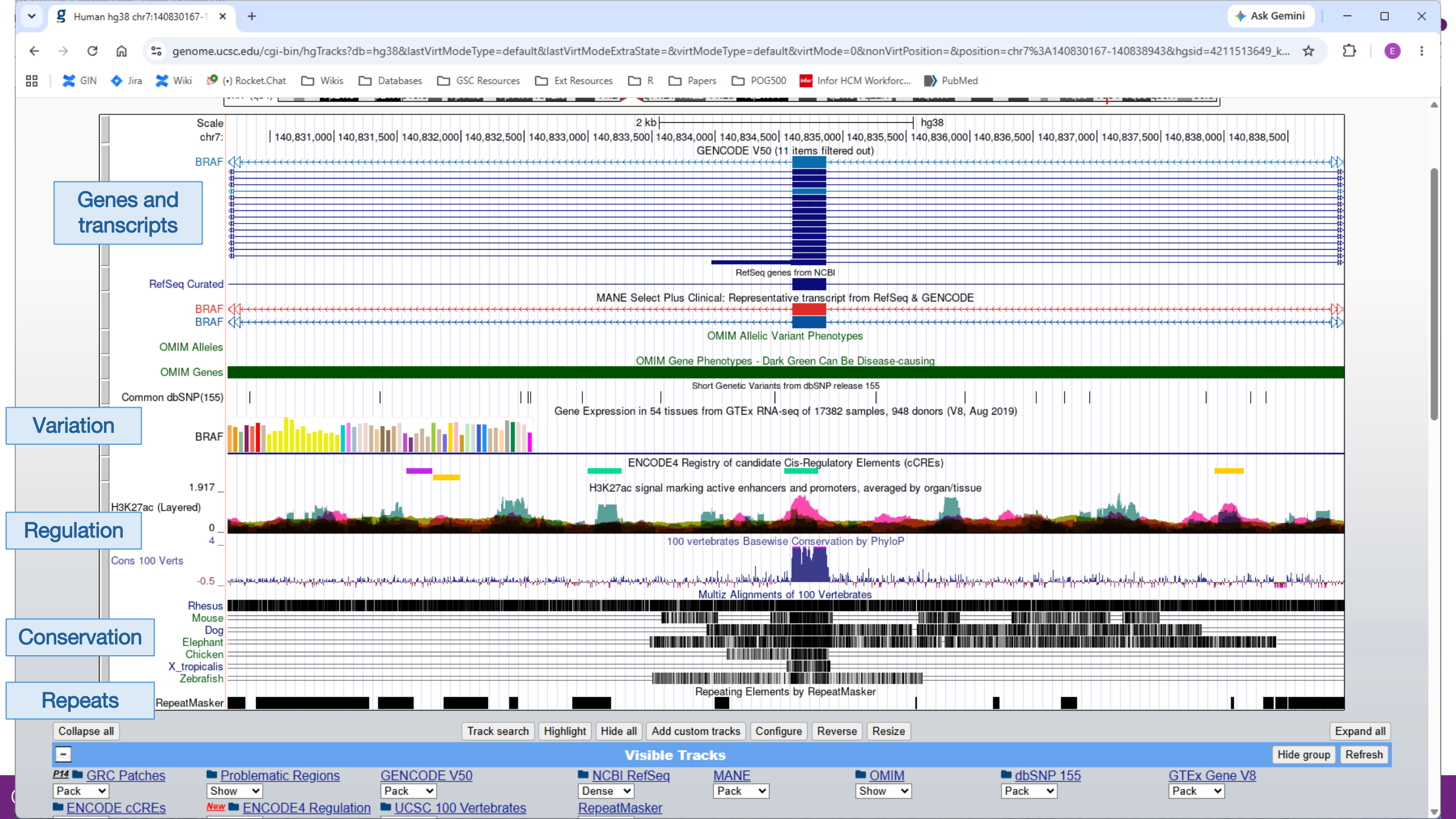
- https://genome.ucsc.edu



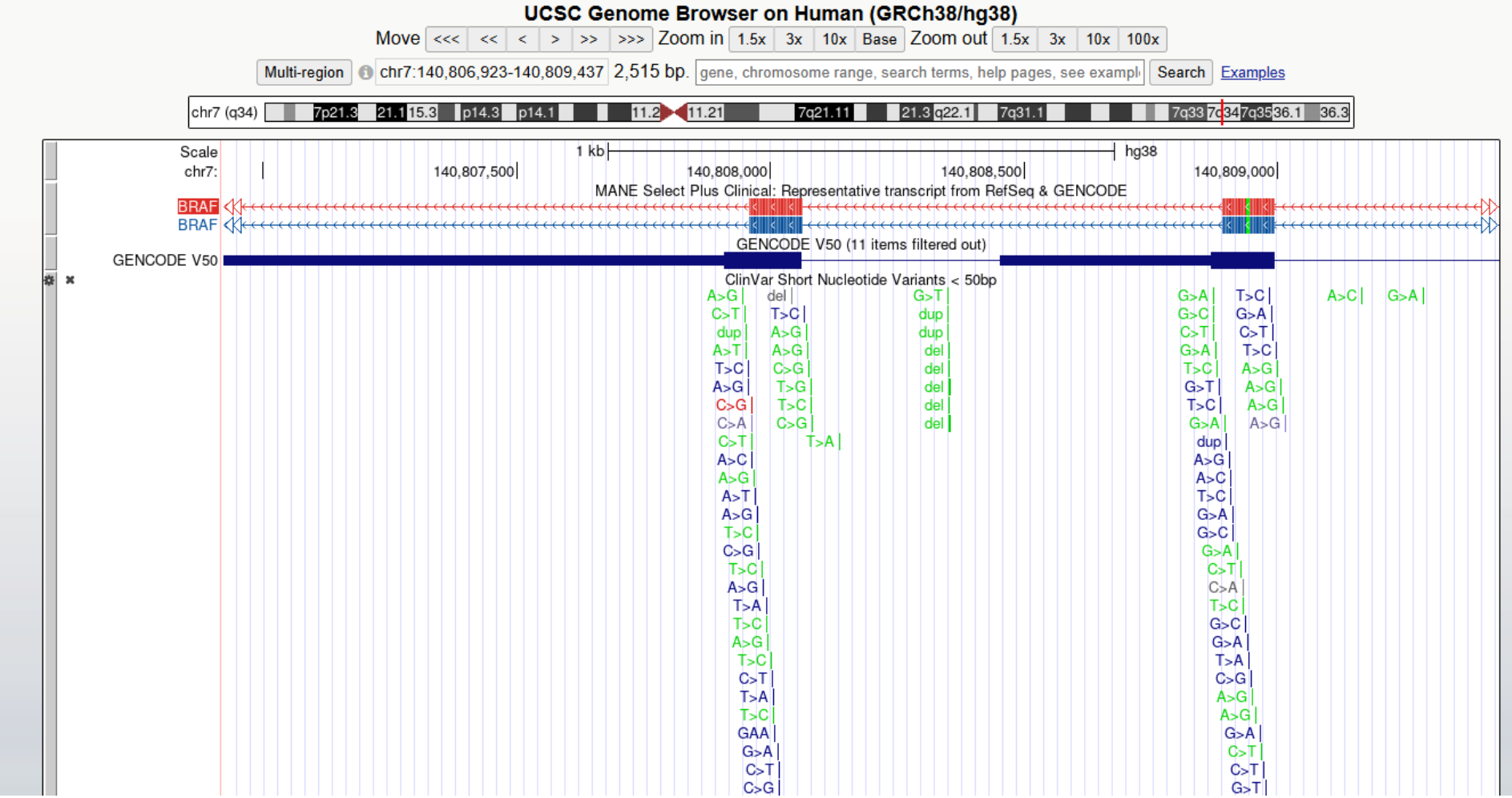
Collapse all | Track search | Highlight | Hide all | Add custom tracks | Configure | Reverse | Resize | Expand all

Visible Tracks | Hide group | Refresh

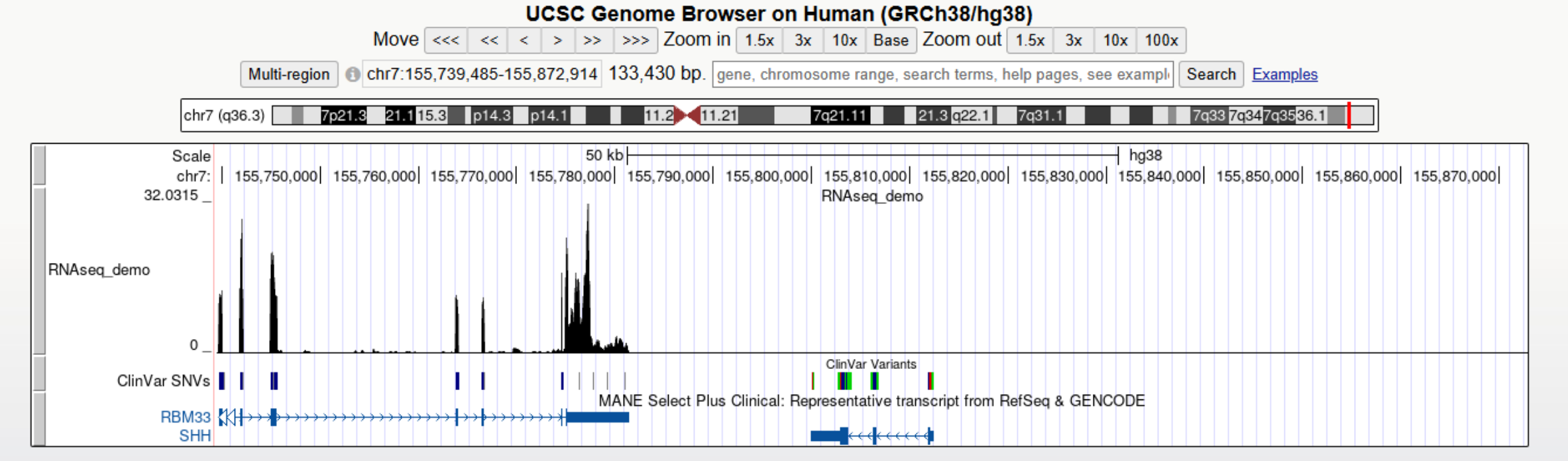
P14 GRC Patches: Pack	Problematic Regions: Show	GENCODE V50: Pack	NCBI RefSeq: Dense	MANE: Pack	OMIM: Show	dbSNP 155: Pack	GTEx Gene V8: Pack
ENCODE cCREs	New ENCODE4 Regulation	UCSC 100 Vertebrates	RepeatMasker				


Genes and transcripts
Variation
Regulation
Conservation
Repeats
Human hg38 chr7:140830167-
Ask Gemini
genome.ucsc.edu/cgi-bin/hgTracks?db=hg38&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr7%3A140830167-140838943&hgsid=4211513649_k...
GIN
Jira
Wiki
(•) Rocket.Chat
Wikis
Databases
GSC Resources
Ext Resources
R
Papers
POG500
Infor HCM Workforc...
PubMed
Scale
chr7:
2 kb
hg38
140,831,000 140,831,500 140,832,000 140,832,500 140,833,000 140,833,500 140,834,000 140,834,500 140,835,000 140,835,500 140,836,000 140,836,500 140,837,000 140,837,500 140,838,000 140,838,500
GENCODE V50 (11 items filtered out)
BRAF
RefSeq genes from NCBI
RefSeq Curated
MANE Select Plus Clinical: Representative transcript from RefSeq & GENCODE
BRAF
BRAF
OMIM Allelic Variant Phenotypes
OMIM Alleles
OMIM Gene Phenotypes - Dark Green Can Be Disease-causing
OMIM Genes
Short Genetic Variants from dbSNP release 155
Common dbSNP(155)
Gene Expression in 54 tissues from GTEx RNA-seq of 17382 samples, 948 donors (V8, Aug 2019)
BRAF
ENCODE4 Registry of candidate Cis-Regulatory Elements (cCREs)
1.917
H3K27ac signal marking active enhancers and promoters, averaged by organ/tissue
H3K27ac (Layered)
0
4
100 vertebrates Basewise Conservation by PhyloP
Cons 100 Verts
-0.5
Multiz Alignments of 100 Vertebrates
Rhesus
Mouse
Dog
Elephant
Chicken
X_tropicalis
Zebrafish
Repeating Elements by RepeatMasker
RepeatMasker
Collapse all
Track search
Highlight
Hide all
Add custom tracks
Configure
Reverse
Resize
Expand all
Visible Tracks
Hide group
Refresh
P14 GRC Patches
Pack
Problematic Regions
Show
GENCODE V50
Pack
NCBI RefSeq
Dense
MANE
Pack
OMIM
Show
dbSNP 155
Pack
GTEx Gene V8
Pack
ENCODE cCREs
New ENCODE4 Regulation
UCSC 100 Vertebrates
RepeatMasker

View variation



Load custom data tracks



- Text: `BED` `bedGraph` `GFF/GTF` `VCF` `WIG`
- Big binary (large data, by URL): `bigBed` `bigWig` `BAM` `VCF+tabix` `.hic`

Table browser

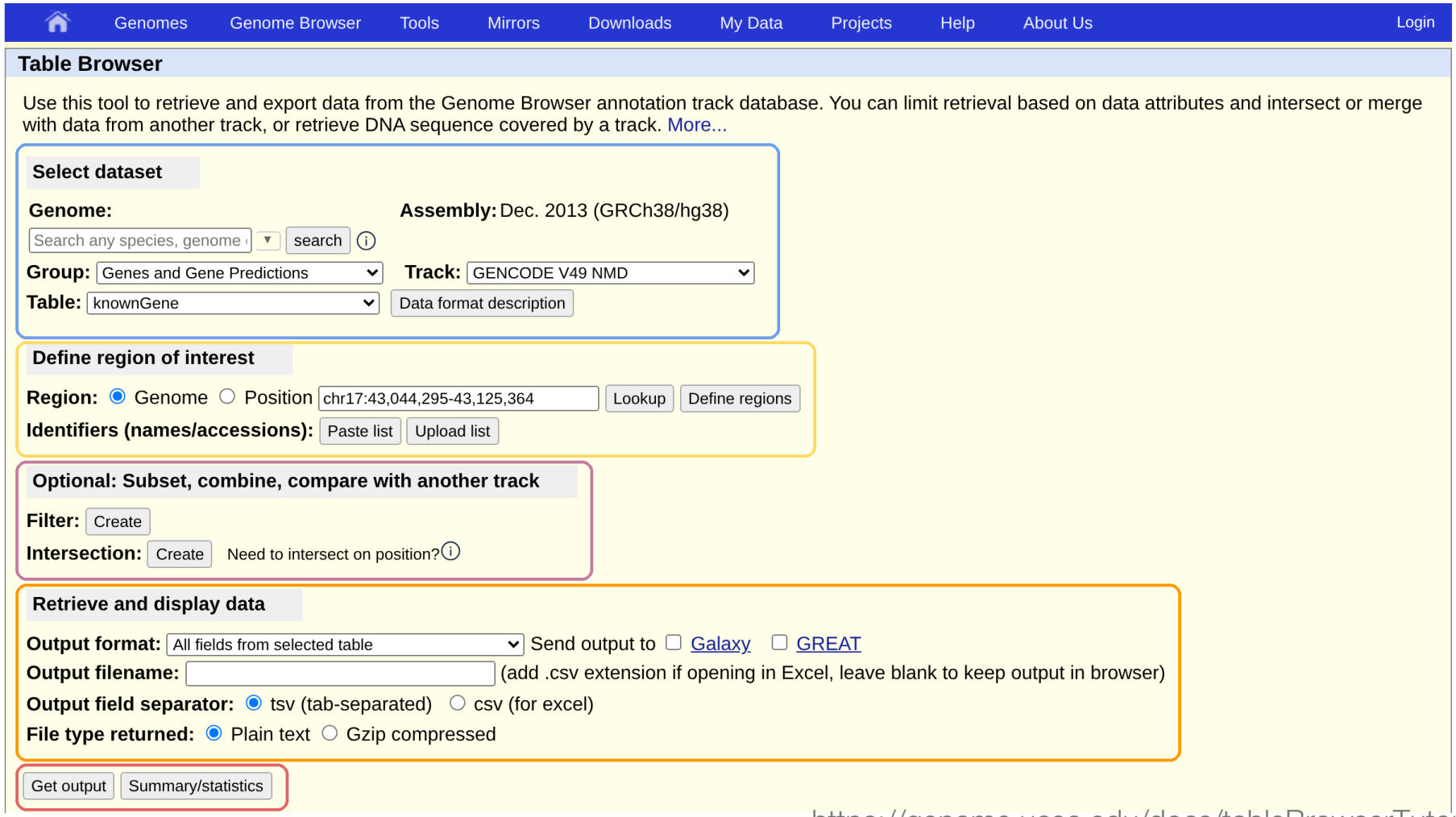
Choose the assembly, track, and table

Select genomic region for data extraction

Create a filter or intersect the table with another table

Choose your output preferences

Execute the selected options



https://genome.ucsc.edu/docs/tableBrowserTutorial.html

Visualization summary

- No single visualization works for every genomic question
- **IGV** for detailed sequencing review
- **UCSC Genome Browser** or Ensembl for public annotations
- **PCA, heatmaps, network diagrams** etc for data analysis

- **Choose the visualization that best answers the biological question**

Coffee Break & Networking Session

Workshop Sponsors